

MOSAIC

Modular Open Source Architecture In Cabin

Technical Specification — Revision 0.11 DRAFT

SEPTEMBER 2026

Revision History

Revision	Date	Author	Change Summary
Rev 0.1	November 14 th , 2025	C. Hall	Initial draft — architecture, VLAN, MQTT, lighting overview, migration from Github
Rev 0.2	April 11 th , 2026	C. Hall A. King	Added WLED REST API, WLED MQTT topics, Thread architecture (§2.3), SPE physical layer (§2.9.9), OTA firmware (§3.4.5), health reporting (§5.2)
Rev 0.X		C. Hall	WLED health alignment (§3.4.6, §5.2–5.4), standard API reference (§5.5), standard WLED commands (§3.4.2 expanded); table header alignment fix (§3.4.2); added backbone/distribution switch power, environmental, and timing requirements (§2.9.9); established AES67 as the primary audio-over-IP transport, with Dante hardware operating in AES67 mode (§6); added Master Server/Router wired interface requirement and flagged a backbone SFP+ conflict against the confirmed router interface set (§2.9.8)
Rev 0.11	September 3 rd , 2026	C. Hall	Refined PTP requirements (§2.9.9, §6.3.1, new §6.3.5); added strict VLAN separation requirements (§2.9.2); established the 10.XX.YY.ZZZ address structure with the optional ATA sub-chapter as the third octet (§2.9.2, §2.9.6, §2.9.7, §3.1) and aligned example addresses throughout; flagged open sub-chapter code assignments (§2.9.6), AES67 DSCP markings (§6.3.2), and PTP grandmaster selection (§6.3.5) for review; made native AES67 hardware preferred over Dante, with Dante accepted subject to the additional network changes of new §6.3.6, and set QoS markings to AES67 defaults (§6.3.2); made tri-band IEEE 802.11ax (2.4/5/6 GHz) the preferred WLAN standard, with a 6 GHz channel plan and the FCC aircraft restriction of 47 CFR §15.407(d)(1)(iv) (§2.9.10); allowed Layer 3 control at the Master Router or a Layer 3 switch (§2.9.2, §2.9.4); split the core network (VLAN 15 / 10.15.0.0/16), general passenger network (VLAN 16 / 10.16.0.0/16), and crew network (VLAN 17 / 10.17.0.0/16), with traffic between core and passenger networks allowed as design and FAA ASISP certification permit (§2.9.3, §2.9.4, Appendix A); assigned all AES67 and Dante devices to VLAN 67 / 10.67.0.0/16 and added a multicast flooding note (§2.9.3, §2.9.6, §6.3.1, §6.6); redrew Figure 2.9.9 on its own page; added §7 Video Distribution (format

		requirement: 4K60 4:4:4 HDR over fiber, IPMX / SMPTE ST 2110 preferred); clarified local versus CMS-called scenes (§3.4.4); moved Component Identification and Labeling to §8 and renumbered Audio Integration to §6 and Video Distribution to §7, with all cross-references updated; added References; added §1.5 Incorporating This Specification; expanded the §2.1 citations in References; added the WLAN white paper cited in §2.9.10 to References; stated the digital-source assumption and added the typical signal chain graphic (§6.2, Figure 6.2.1); added the interface preference chart (Figure 2.2.1); added a note discouraging RS-485 and similar serial data buses for inter-system communication (§2.2.1) and a placeholder §9 MOSAIC Open Architecture Display (MOAD); stated that MQTT is not mandatory and aligned MQTT requirements accordingly (§2.2.5, §2.2.6, §3.3, §3.4.5, §5.2–5.4); reduced §6.5 to a note that AES67 audio is not intended for certified passenger address
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1. Overview

1.1 What Is MOSAIC?

MOSAIC (Modular Open Source Architecture in Cabin) is a framework designed to enhance the modularity and interoperability of aircraft cabin systems. The core fundamentals of MOSAIC are derived from the Modular Open Systems Approach (MOSA) used in military environments, which emphasizes flexibility, upgradeability, and cost-effectiveness.

1.2 Key Attributes

Interoperability

Enables systems and software applications to access and provide data and services using open interface definitions between components. This ensures different systems can communicate and exchange data seamlessly — for example, an in-flight entertainment system integrating with various seat configurations and passenger service units.

Tech Refresh

Enables periodic upgrades of technology to ensure system supportability and performance. For instance, upgrading cabin lighting to more energy-efficient LED technology without requiring a complete system overhaul.

Modularity

Enforces the establishment of well-defined, standardized system modules that can be tested individually for function and conformance. [SOSA 3.1]

Innovation

Uses business practices that encourage the relevant technical community to develop and insert new technologies, fostering a culture of continuous improvement. An example is enabling third-party developers to create new applications for the cabin management system.

Competition

Prevents vendor lock-in and increases options for component replacement and refresh, promoting a competitive market with better pricing and greater supplier diversity.

Cost Savings / Cost Avoidance

Reduces the total cost of ownership by minimizing maintenance and upgrade costs. For example, a modular seat design that allows replacement of individual components rather than the entire assembly.

1.3 The Need for Modularity in Aircraft Cabins

Attribute	Benefit
Flexibility	Modular designs allow easy reconfiguration of cabin layouts to meet different airline requirements and passenger preferences.
Upgradeability	Modular systems can be upgraded incrementally, reducing downtime and costs associated with major overhauls.
Cost-Effectiveness	Standardized interfaces and components reduce inventory costs and simplify maintenance procedures.
Passenger Experience	Modular systems enable integration of the latest technologies and amenities, improving overall comfort and satisfaction.

1.4 Vendor Involvement

- Vendor involvement is required.
- Vendors should include one or more of the defined interfaces into products where practical.
- Software command structures should be posted and free to use without license restrictions (e.g., custom API commands to control a display).
- MOSAIC technical and business architecture is vendor-agnostic. [SOSA 3.2.1]
- MOSAIC is designed to protect and not expose vendor intellectual property. [SOSA 3.2.1]

1.5 Incorporating This Specification

This section defines how a vendor incorporates MOSAIC requirements into a product, modification, or installation.

- **Partial adoption.** This specification is not intended to be applied only as a whole. It may be applied in part, depending on the scope of the modification or installation. For example, MQTT and Thread are recommendations, not hard requirements; MQTT is one means of communication, not the only means.
- **Review of changes.** Changes and updates to this specification are reviewed by the MOSAIC Working Group before they are incorporated.
- **Published communication.** Vendor-specific communication between systems shall be openly published to the MOSAIC repository (GitHub or equivalent). Customizations or forks of open-source commands or software should be identified and defined.
- **Proprietary internals.** Proprietary commands and software internal to a subsystem are not required to be published, provided that (a) the tools needed to troubleshoot the subsystem are freely available or published, and (b) they apply only within the subsystem itself and not to communication with external systems.
- **Deviations.** This specification is a framework. Where a deviation is needed to support new technology integration or a new system design, the MOSAIC Working Group should be notified so that this specification can be updated if needed.

2. MOSAIC Framework

2.1 Architecture

Modular architecture is defined as a system architecture where the system is partitioned into components that exhibit good abstraction, high internal cohesion, low coupling to other components and external systems, and encapsulate their internal implementation behind well-defined interfaces (Firesmith and Donohoe 2015).

Modularity is a set of principles for managing complexity by breaking up a complex system into discrete pieces that communicate through standardized interfaces, eliminating unmanageable interconnections (Langlois 2002).

Implementation steps include:

- Modularize the system by decomposing it into smaller, manageable modules, each with a clear and distinct function.
- Specify or configure interfaces to define how modules interact, ensuring interfaces are standardized and well-documented.
- Define interface specifications — including data formats, communication protocols, and error handling mechanisms.
- Standardize interface specifications so all modules adhere to the defined standards, promoting consistency and interoperability.
- Require open interfaces at the boundaries of each module to ensure interoperability and facilitate integration with third-party components.

2.2 Interface Standards

2.2.1 Interface Types

Interface Type	Description
Analog	Continuous signal communication (voltage/current levels). See §2.2.2.
Physical Discrete	Basic digital I/O between systems. Minimum required interface. §2.2.3.
Virtual Discrete	Software/timer-calculated discrete events. §2.2.4.
Publish-Subscribe	One-to-many messaging via MQTT broker. §2.2.5.
Request-Response	Bidirectional query/reply using MQTT or HTTP/REST. §2.2.6.
Event Notification	Asynchronous event-driven notification. §2.2.7 — TBD.
Point-to-Point	Direct unicast communication between two endpoints. §2.2.8 — TBD.

The interface types are ranked by preference as shown in Figure 2.2.1. A network API is the most preferred method of interface. Physical discrete and analog are the lowest tier and represent the minimum.

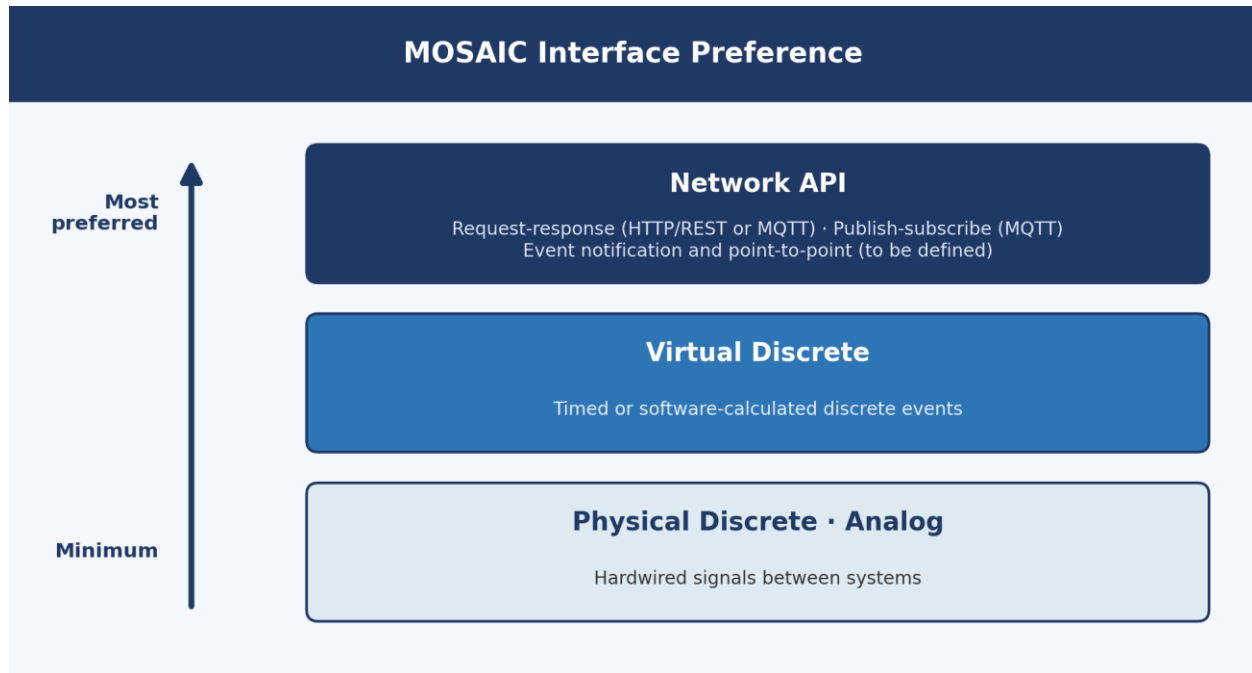


Figure 2.2.1 — MOSAIC interface preference. A network API is the most preferred method of interface between systems; physical discrete and analog are the minimum.

Serial Data Buses (RS-485 and Similar)

Serial data buses such as RS-485, and similar types such as RS-232 and RS-422, are discouraged for communication between systems. The reasons are:

- **Unpublished protocols.** RS-485 defines only the electrical connection. The command set carried on the bus is defined by each vendor and is often not published.
- **Version differences.** Where a command set is available, different versions of the same system often use different commands, so documentation for one version cannot be relied on for another.
- **Checksums.** Messages normally carry a vendor-specific checksum. Unless the checksum method is published, messages cannot be generated or verified independently.
- **Limited test tools.** Few tools are available that let an aviation maintenance technician without specialist training observe, test, and understand bus traffic accurately, which makes faults difficult to isolate.

Serial data buses remain acceptable within a subsystem. Where one is used between systems, the publication requirements of §1.5 apply.

2.2.2 Analog

Analog interface specifications to be defined in a future revision. This section is a placeholder for continuous signal definitions including voltage range, impedance, and sampling requirements.

2.2.3 Physical Discrete

- Basic input/output between systems — the minimum required interface.
- Can be active low (ground) or active high (>22 VDC); active high preferred.

2.2.4 Virtual Discrete

- Based on timed or software-calculated events.
- Can be active low (ground) or active high (28 VDC).

2.2.5 Publish-Subscribe (MQTT)

One publisher of messages to one or more subscribers per topic. MQTT is the preferred communication protocol.

MQTT is not mandatory. It is one means of communication between systems, not the only means. Requirements in this specification that reference MQTT apply where MQTT is implemented; where it is not, the equivalent function is provided through another interface defined in §2.2, such as HTTP/REST or physical discrete.

MQTT Broker Requirements

- Broker software must be open source (e.g., Eclipse Mosquitto or EMQX).
- Broker must reside on the Master Server/Router (see §2.9.2).
- TLS 1.2 or higher is required for all broker communications.
- Client authentication via username/password or X.509 certificate is required.
- Retained messages must be enabled for system-state topics.
- Last Will and Testament (LWT) must be configured for all critical LRUs to broadcast a disconnect event on unplanned disconnection.

Quality of Service (QoS)

QoS Level	Name	Recommended Use
0	At Most Once	Non-critical telemetry; sensor streaming where occasional loss is acceptable.
1	At Least Once	Commands, state changes, lighting control. Default for cabin systems.
2	Exactly Once	Financial transactions, passenger manifests, safety-critical state.

Topic Naming Convention

Topics must follow the hierarchical structure below using forward-slash delimiters:

```
mosaic/<ata>/<subsystem>/<lru-id>/<function>
```

Examples:

```
mosaic/33/lighting/row-12a/brightness
mosaic/44/cms/screen-01/power-state
mosaic/25/seat/seat-14b/recline-position
```

- ATA chapter numbers align with the IP addressing scheme defined in §2.9.
- LRU IDs must match the device's assigned network address suffix.

- Function names must be lowercase, hyphen-separated, and documented in the LRU's interface manifest.

Retained Topics

The following topic categories must be published with the MQTT retained flag enabled so newly connected subscribers immediately receive current state:

- Power state of all LRUs
- Health/fault status (see §5.0)
- Lighting zone level and color values
- Cabin mode (e.g., boarding, cruise, dining, landing)

2.2.6 Request-Response

There is no single preferred protocol for request-response. MQTT and HTTP/REST are both acceptable; neither is mandatory. REST endpoints are defined in §2.9.11.

2.2.7 Event Notification

Specification to be defined in a future revision. Intended for asynchronous push notifications from subsystems to monitoring agents without requiring a polling loop.

2.2.8 Point-to-Point

Specification to be defined in a future revision. Intended for direct unicast sessions between two known endpoints where pub/sub overhead is undesirable.

2.3 Thread Network (IoT Mesh)

Thread is a low-power, IPv6-based mesh networking protocol designed for reliable IoT device communication. It is the preferred wireless protocol for distributed cabin sensors and seat electronics that cannot support full Ethernet connectivity. All lighting controllers use wired Ethernet and do not join the Thread mesh.

2.3.1 Thread Architecture

- Thread operates as a self-healing mesh over IEEE 802.15.4 radio (2.4 GHz).
- All Thread nodes form a mesh and automatically route around failed nodes.
- A Thread Border Router (TBR) provides the bridge between the Thread mesh and the cabin LAN (§2.9.3).
- The TBR must be collocated with or integrated into the Master Server/Router (§2.9.2).
- Multiple TBRs may be deployed for redundancy in large cabin configurations.

2.3.2 Thread Node Roles

Role	Description
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Border Router	Bridges the Thread mesh to the cabin IP/LAN. Hosts the MQTT broker relay.
Router	Full-capability node; routes traffic within the mesh. Mains-powered preferred.
End Device	Leaf node; communicates only with its parent router. Suitable for battery/seat-power nodes.
Sleepy End Device	Low-power variant that polls its parent at intervals. Use for infrequently sampled sensors.

2.3.3 Thread and MQTT Integration

Thread nodes communicate to the MQTT broker via the Thread Border Router. The TBR performs NAT64 or direct IPv6 forwarding to the broker on the cabin LAN. MOSAIC lighting controllers connect via wired Ethernet (see §3.4) and do not participate in the Thread mesh. Thread nodes that carry sensor or seat data publish to MQTT via this path.

2.3.4 Thread Commissioning

- All Thread devices must be pre-commissioned before installation using a commissioning tool or QR-code out-of-band method.
- Network credentials (PAN ID, Extended PAN ID, network key) must be unique per aircraft and stored in the Master Server.
- The Matter protocol (CSA) should be evaluated as a commissioning and interoperability layer above Thread for future hardware revisions.

2.9 IP-Based Architecture

2.9.1 Design Goals

The MOSAIC IP architecture is built around three requirements that must be satisfied simultaneously:

Goal	Description
Subsystem Isolation	Each ATA chapter is a distinct VLAN and broadcast domain. A failure or compromise in one VLAN does not propagate directly to others.
Controlled Inter-VLAN Routing	Explicit firewall rules permit cross-VLAN communication only for defined system pairs. Example: Lighting (VLAN 33) can receive commands from the CMS (VLAN 44) and a user device on VLAN 16 or VLAN 17, but cannot initiate connections to power or fire-protection VLANs.
WAN Isolation	No MOSAIC subsystem has a routable path to external networks. No WAN interface is configured on the Master Router. SATCOM and passenger internet traffic is carried on a physically separate network.

2.9.2 VLAN Architecture — One VLAN per ATA Chapter

Each ATA chapter is assigned one IEEE 802.1Q VLAN and one /16 IPv4 subnet. The VLAN ID matches the ATA chapter number. All subsystems within a chapter share a broadcast domain and are differentiated by IP address only. The /16 block provides up to 65,534 host addresses per VLAN, eliminating device count limits for any foreseeable cabin deployment. Inter-VLAN traffic is routed by the Layer 3 control point — the Master Router or a Layer 3 switch, depending on the program design — with firewall policy enforced at each VLAN interface.

Every MOSAIC device address follows the structure 10.XX.YY.ZZZ, where XX is the ATA chapter (major code, identical to the VLAN ID), YY is the ATA sub-chapter (minor code) identifying the subsystem, and ZZZ is the device ID (1–254) assigned at commissioning (§2.9.7). Example: Wash lighting, ATA 33-10, is addressed 10.33.10.ZZZ. YY = 0 is reserved for chapter-level infrastructure, including the VLAN gateway at 10.XX.0.1.

NOTE: All lighting controllers — regardless of zone — are on VLAN 33 / 10.33.0.0/16. The ATA sub-chapter (YY) octet is optional in every device address, but it has no routing significance: all YY blocks within a chapter share one VLAN and one /16 subnet.

Strict VLAN Separation

VLAN separation is enforced at Layer 2 on every switch port and at Layer 3 on the Layer 3 control point, which is the Master Router or a Layer 3 switch depending on the program design. The following requirements apply to all MOSAIC switches, the Master Router, and all LRUs:

- Access ports: each LRU-facing port is an untagged member of exactly one VLAN — the VLAN of the connected LRU's ATA chapter. 802.1Q-tagged frames received on an access port must be dropped.
- Trunk ports: each trunk carries only an explicitly configured list of the MOSAIC VLANs required on that link. All trunk traffic is 802.1Q-tagged; untagged frames received on a trunk port must be dropped.
- VLAN 1 and any switch default or native VLAN must not carry MOSAIC traffic and must be removed from all access and trunk ports.
- Ingress filtering must be enabled on every port so that frames tagged for a VLAN the port is not a member of are dropped.
- VLAN membership is statically configured. Dynamic VLAN registration (GVRP/MVRP) and automatic trunk negotiation must be disabled.
- Unused switch ports must be administratively disabled.
- An LRU connects to one VLAN only. No LRU may bridge, route, or proxy traffic between VLANs, and no switch may do so unless it is the designated Layer 3 control point; the Layer 3 control point is the only inter-VLAN path, subject to the firewall policy of §2.9.4.
- Source-address validation: the Layer 3 control point must drop any packet received on a VLAN interface whose source address is outside that VLAN's 10.XX.0.0/16 subnet.
- Broadcast and multicast traffic — including AES67 streams and PTP messages (§6.3.5) — is confined to its originating VLAN. Multicast routing, IGMP proxying, and PTP forwarding between VLANs must not be enabled on the Layer 3 control point.

2.9.3 VLAN Address Structure

VLAN ID	Subnet	Gateway	Description
15	10.15.0.0/16	10.15.0.1	Core Network — Master Router, MQTT Broker, NTP, DHCP
16	10.16.0.0/16	10.16.0.1	General Passenger Network Access — passenger devices, Passenger Wi-Fi
17	10.17.0.0/16	10.17.0.1	Crew Network Access — crew devices, Crew Wi-Fi
21	10.21.0.0/16	10.21.0.1	Pneumatics
23	10.23.0.0/16	10.23.0.1	Communications — PA Systems, SATCOM uplink (physically separate)
24	10.24.0.0/16	10.24.0.1	Power Systems
25	10.25.0.0/16	10.25.0.1	Galley Equipment and Seats
26	10.26.0.0/16	10.26.0.1	Fire Protection (read-only from other VLANs)
33	10.33.0.0/16	10.33.0.1	All Lighting — area, wash, reading, toe kick, task, RGB
44	10.44.0.0/16	10.44.0.1	CMS / IFE — core, map, camera, audio, video, disc I/O
67	10.67.0.0/16	10.67.0.1	AES67 / Dante Audio — all audio-over-IP devices and PTP

⚠ NOTE: VLAN 15 / 10.15.0.0/16 is the core network: Master Router, MQTT broker, NTP, DHCP server, and other management infrastructure. VLAN 16 / 10.16.0.0/16 is the general passenger network for passenger devices. VLAN 17 / 10.17.0.0/16 is the crew network for crew devices. Inter-VLAN traffic between VLAN 15 and VLAN 16 is allowed as the program design and its FAA Aircraft Systems Information Security/Protection (ASISP) certification permit. Firewall INPUT rules on the VLAN 16 and VLAN 17 interfaces must restrict passenger and crew device access to the router itself to DHCP (UDP 67), DNS (UDP/TCP 53), and NTP (UDP 123) only — passenger and crew devices must not reach the router management interface (SSH, HTTP/LuCI). See §A.3 and §A.4 for implementation.

2.9.4 Inter-VLAN Routing and Firewall Policy

The Layer 3 control point (§2.9.2) — the Master Router or a Layer 3 switch, depending on the program design — performs all inter-VLAN routing. There is no default route — any packet with a destination outside the defined VLANs is dropped. Firewall zones map one-to-one to VLANs. All unspecified inter-zone traffic is denied by default.

Source VLAN	Destination VLAN	Policy	Rationale
VLAN 33 (Lighting)	VLAN 33 (Lighting)	FREE — L2	All lighting devices on the same VLAN communicate without routing

VLAN 44 (CMS)	VLAN 33 (Lighting)	PERMIT	CMS sends cabin mode and scene commands to lighting
VLAN 33 (Lighting)	VLAN 44 (CMS)	PERMIT	Lighting reports health and state to CMS via MQTT
VLAN 16 (Passenger), VLAN 17 (Crew)	VLAN 44 (CMS)	PERMIT	Passenger and crew devices access IFE interface
VLAN 16 (Passenger), VLAN 17 (Crew)	VLAN 33 (Lighting)	PERMIT TCP 80,443,1883	User devices may send LAN API commands to lighting (REST/MQTT)
VLAN 15 (Core)	All VLANs	PERMIT	Management, DHCP, NTP, and MQTT broker traffic from the core network
VLAN 16 (Passenger)	VLAN 15 (Core)	PERMIT — per ASISP	Traffic between the passenger and core networks, in either direction, is allowed as the program design and FAA ASISP certification permit
Any	VLAN 26 (Fire)	GET only — port 80	Read-only status monitoring; write commands blocked at firewall
Any	0.0.0.0/0 (WAN)	DENY — dropped	No WAN interface exists; no route to external networks

2.9.5 WAN Isolation Policy

WAN isolation is enforced at the Master Router for all MOSAIC deployments.

Rule	Implementation
No default route	The Master Router must NOT have a 0.0.0.0/0 default route pointing outward. Any unmatched destination is dropped.
WAN interface	TBD
Passenger internet separation	SATCOM and passenger internet traffic must be carried on a physically separate network — never bridged to the MOSAIC cabin network.
Maintenance laptop isolation	External maintenance devices connect via a dedicated management port with firewall access restricted to VLAN 15 only.
Egress monitoring	The Master Server logs any packet dropped due to no matching route or firewall rule. These events are Priority 3 faults (§5.0).

2.9.6 ATA Subsystem Address Reference

The following table documents all MOSAIC VLAN assignments. Each ATA chapter maps to one VLAN and one /16 subnet. All devices within an ATA chapter share the chapter's subnet regardless of subsystem; within the subnet, the third octet (YY) identifies the ATA sub-chapter per §2.9.2.

VLAN	ATA Chapter	Subnet	Description
15	15	10.15.0.0/16	Core Network — Master Router, MQTT Broker, NTP, DHCP
16	16	10.16.0.0/16	General Passenger Network Access — passenger devices, Passenger Wi-Fi
17	17	10.17.0.0/16	Crew Network Access — crew devices, Crew Wi-Fi
21	21	10.21.0.0/16	Pneumatics
23	23	10.23.0.0/16	Communications — PA Systems, SATCOM (uplink physically separate)
24	24	10.24.0.0/16	Power Systems
25	25	10.25.0.0/16	Galley Equipment and Seats
26	26	10.26.0.0/16	Fire Protection — read-only access from other VLANs
33	33	10.33.0.0/16	All Lighting — area, wash, table, reading, toe kick, task, closet, RGB effects
44	44	10.44.0.0/16	CMS / IFE — core, map, camera, audio, video distribution, disc I/O
67	—	10.67.0.0/16	AES67 / Dante Audio — all audio-over-IP devices and PTP (VLAN ID taken from AES67; not an ATA chapter)

⚠ REVIEW REQUIRED: ATA sub-chapter (YY) codes are confirmed only for wash lighting (33-10 → 10.33.10.ZZZ). YY codes for every other subsystem in every chapter must be assigned before device addresses, DHCP reservations, and labels can be finalized. Address ranges shown as 10.33.YY.ZZZ in §3.1 are pending these assignments.

2.9.7 LRU Addressing

- Preferred method: GUI-based address assignment via the Master Server interface.
- Secondary method: External rotary dial or DIP switch on the LRU.
- Discrete wiring for IP ID assignment is accepted as a fallback only.
- Each LRU ships with its ATA Master address hardcoded (e.g., 10.33.10.000 for wash lighting).
- The project-specific device ID (last octet) is assigned during installation commissioning (e.g., 10.33.10.103).
- Static DHCP reservations via MAC address binding are the preferred commissioning method for all wired Ethernet-connected devices including WLED controllers.

⚠ NOTE: Discrete-wiring IP assignment limits the addressable range to the number of I/O lines available and complicates re-addressing. GUI-based or static DHCP assignments are strongly preferred for all new deployments.

2.9.8 Master Server / Router

One dedicated master router serves the entire cabin. It is responsible for overall address management, inter-subsystem routing, MQTT brokering, WAN null routing, and system health reporting.

Minimum Requirements

- System event logging and reporting
- LRU health status monitoring
- System interconnect status
- Open-source software base
- Static routing table with WAN null route (0.0.0.0/0 → DROP)
- ACL enforcement for inter-VLAN firewall policies per §2.9.4
- Minimum of three (3) 2.5GBASE-T and three (3) 1000BASE-T wired Ethernet interfaces

Preferred Requirements

- VRRP with defined failover timer (recommended: 3-second dead interval)
- Over-the-air event status and logging (MQTT, Thread Border Router relay)
- Cybersecurity anomaly detection and reporting
- Layer 4 network management and traffic shaping
- Application virtualization / containerization (e.g., Docker)

2.9.9 Local Area Network (LAN)

The MOSAIC LAN uses a tiered physical layer: a 10G fiber backbone between the Master Server and distribution points, 2.5G copper for zone-level distribution, and Single Pair Ethernet (SPE) as an optional last-mile connection to end nodes such as lighting controllers and seat modules.

Backbone — 10G Fiber

- 10GBASE-SR or 10GBASE-LR fiber between the Master Server/Router and any distribution switches.
- SFP+ ports required on the Master Server and distribution switch uplinks.
- Compact aviation-grade managed switches with SFP+ support are preferred for weight and size constraints.

Distribution — 2.5G Copper

- 2.5GBASE-T (IEEE 802.3bz) copper links from distribution switches to access-layer switch ports.
- Cat6 or Cat6A cable minimum for 2.5G runs.
- Compact managed switches with 2.5G RJ45 ports provide per-VLAN access for devices that cannot use SPE.

- All distribution switches must support 802.1Q VLAN tagging and carry the appropriate MOSAIC VLANs as tagged trunk ports.

End Nodes — Single Pair Ethernet (SPE) Option

Single Pair Ethernet is an optional physical medium for end nodes — specifically lighting controllers, sensors, and seat modules — where cable weight, space, or run length make 8-conductor copper impractical.

Standard	Speed / Range	Recommended Use
10BASE-T1L (IEEE 802.3cg)	10 Mbps / up to 1000 m	Lighting controllers, seat sensors, distributed I/O — long runs over a single twisted pair
100BASE-T1 (IEEE 802.3bw)	100 Mbps / up to 15 m	Short-range end nodes requiring higher throughput; common in automotive-derived hardware

- SPE uses a single twisted pair versus the four pairs of standard Cat5/6, significantly reducing cable harness weight over long cabin runs.
- SPE supports Power over Data Line (PoDL, IEEE 802.3bu) — a single cable carries both data and 12–52 VDC power to the end node, eliminating a separate power conductor.
- An SPE-to-standard-Ethernet media converter or SPE-capable managed switch port is required at the distribution layer. The network segment above the media converter operates at standard Ethernet speeds.
- End nodes using SPE must include an SPE PHY (physical layer transceiver) compatible with 10BASE-T1L. WLED-based controllers with native RJ45 ports require an SPE-to-standard-Ethernet media converter at the SPE segment termination where 8-conductor copper is not viable.

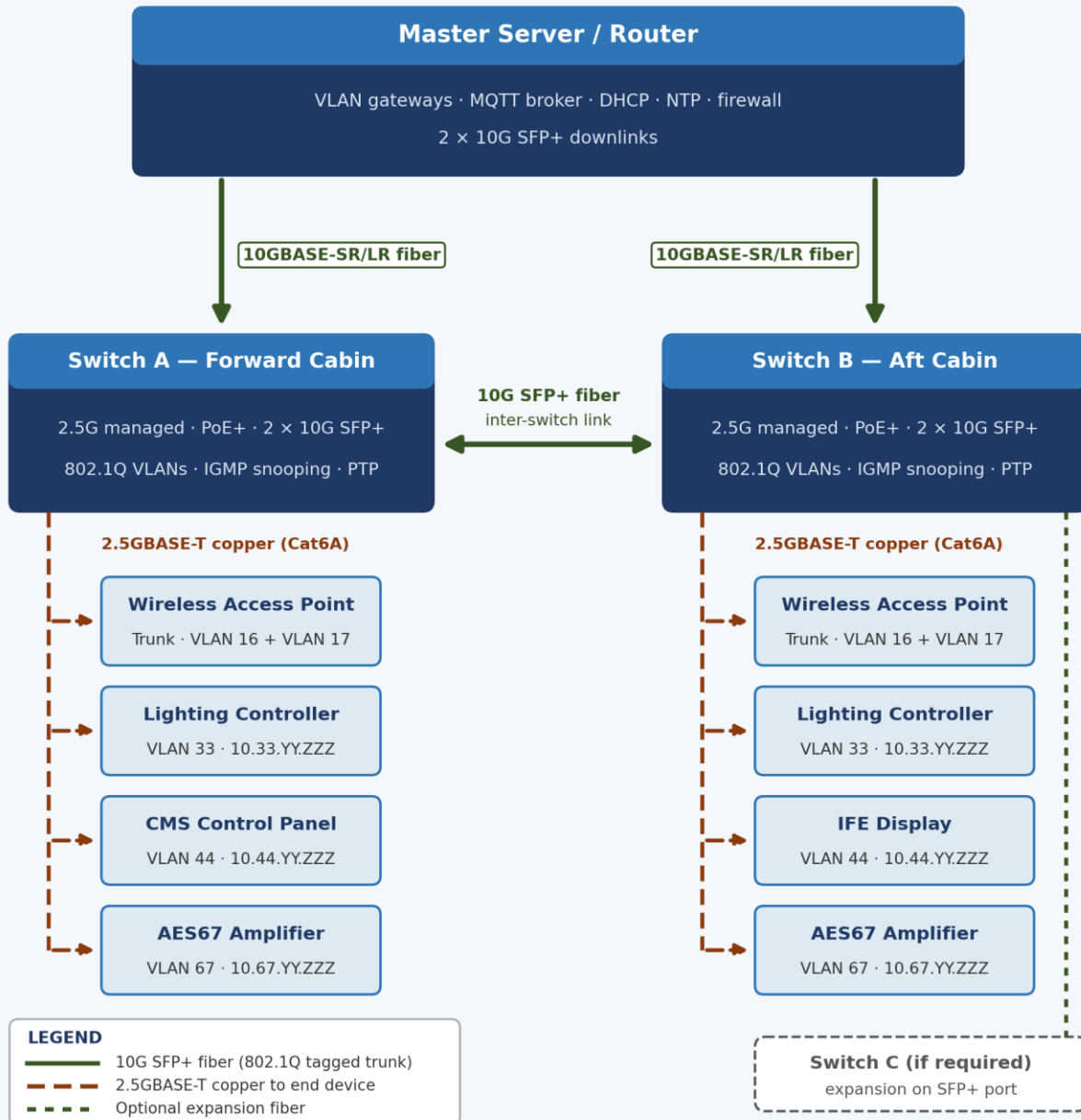
NOTE: SPE is well established in industrial and automotive networking. Reference IEC 63171-6 or equivalent.

Physical Layer Summary

Tier	Medium	Speed	Typical Use
Backbone	SFP+ Fiber	10 Gbps	Master Server ↔ Distribution Switch
Distribution	Cat6/6A Copper	2.5 Gbps	Distribution Switch ↔ Access Switch
End Node (standard)	8-conductor Cat5e/6	100/1000 Mbps	CMS, AP, IFE screens, media players
End Node (SPE)	Single Twisted Pair	10/100 Mbps + PoDL	Lighting controllers, seat sensors, I/O modules

MOSAIC LAN — Dual-Distribution-Switch Topology

Minimum configuration: two managed switches on a 10G fiber backbone



Device ports are untagged access ports in one VLAN; access point ports are trunks limited to VLAN 16 and VLAN 17 (§2.9.2).

A device keeps its VLAN and 10.XX.YY.ZZZ address regardless of which switch it connects to.

Figure 2.9.9 — MOSAIC LAN dual-distribution-switch topology (minimum configuration). Master Server connects to each switch via 10G SFP+ fiber. An additional 10G SFP+ inter-switch link (ISL) interconnects Switch A (forward cabin) and Switch B (aft cabin). End devices connect via 2.5GBASE-T copper as untagged access ports in their subsystem VLAN; wireless access point ports are trunks carrying VLAN 16 and VLAN 17. Switch B carries an expansion SFP+ port for daisy-chaining a Switch C if cabin length or device count requires it.

Switch Power, Environmental, and Timing Requirements

All backbone and distribution switches selected for the MOSAIC LAN must meet the following requirements. These requirements ensure reliable operation from the aircraft 28 VDC bus, sealed installation in cabin and unconditioned-bay locations, and the switch-fabric timing and multicast behavior needed to support AES67 audio distribution (§6.3).

Requirement	Specification
Power Input	Redundant DC input across a minimum 24–55 VDC range, sourcing both switch logic and all PoE-enabled ports directly from the aircraft 28 VDC bus without an external step-up/boost converter. Reverse-polarity and overload protection required on each input.
PoE Output	IEEE 802.3af/at PoE+ on all downstream copper ports (≥ 30 W/port). The switch's total PoE power budget must be published and validated against the sum of connected loads; port priority or power-management scheduling must be configured on any switch where aggregate port demand can exceed the chassis PoE budget.
Fiber Uplink	Minimum of two 1G/10G SFP+ uplink ports for backbone connections per Backbone — 10G Fiber, above.
Sealing / Connectors	IP67-rated sealed metal enclosure with M12 connectors (X-coded for data, K-coded for power) for switches mounted in cabin or unconditioned bay locations.
Environmental	Extended operating temperature range suitable for the installation location; -40°C to $+70^{\circ}\text{C}$ or better is preferred for unconditioned bays.
Ring Redundancy	ITU-T G.8032 (ERPS) or an equivalent standards-based ring protocol providing sub-50 ms recovery, in addition to STP/RSTP/MSTP.
Time Synchronization	IEEE 1588-2008 (PTPv2) with hardware timestamping, operating as a boundary clock (preferred) or end-to-end transparent clock on all ports carrying the audio VLAN, to preserve timing accuracy for AES67 audio distribution across the switch fabric. PTP must be disabled on all other ports. See §6.3.5.
Multicast Handling	IGMP snooping v2/v3 to prevent flooding of AES67 audio streams and other multicast control traffic to non-subscribed ports.
Compliance	TAA/NDAA/BABA-compliant sourcing per program procurement requirements, with an open, non-proprietary management stack consistent with §1.4.

2.9.10 Wireless Area Network (WLAN) — Passenger and Crew Devices Only

The MOSAIC WLAN is provided exclusively for passenger devices on VLAN 16 and crew tablets on VLAN 17. All lighting controllers, sensors, and other LRUs connect to VLAN 33 via wired Ethernet (§2.9.9). No lighting controller or ATA 33 device may connect to the cabin network over Wi-Fi. The requirements below apply to WLAN access point hardware and passenger/crew device connectivity only.

Minimum Requirements

- Gigabit (10/100/1000) wired uplink interfaces on all access points.
- IEEE 802.11 with 2.4 GHz and 5 GHz channel support.
- 2.4 GHz: Channels limited to 1, 6, and 11 (non-overlapping) only.
- 5 GHz: 40 MHz channel spacing minimum.
- 6 GHz band support is optional at the minimum level (see Preferred Requirements).

Preferred Requirements

- IEEE 802.11ax (Wi-Fi 6E) access points with simultaneous operation on all three bands: 2.4 GHz, 5 GHz, and 6 GHz.
- 5 GHz: 80 MHz channel spacing, reverting to 40 MHz in high-interference environments.
- 6 GHz: Operation limited to U-NII-5 (5.925–6.425 GHz) only; see the FCC note below.
- 6 GHz: 80 MHz channel spacing using the six non-overlapping U-NII-5 channels (channel centers 7, 23, 39, 55, 71, and 87), reverting to 40 MHz in high-interference environments. 160 MHz channels are not used.
- 6 GHz: Primary 20 MHz channel set to a Preferred Scanning Channel (5, 21, 37, 53, 69, or 85) so that client devices discover the network.
- 6 GHz: WPA3 (or Enhanced Open) is required on every SSID broadcast in this band; WPA2-only SSIDs remain on 2.4 GHz and 5 GHz.
- Multi-SSID with per-SSID bandwidth allocation (crew vs. passenger isolation).
- WPA3-Enterprise authentication for all cabin management SSIDs.

⚠ NOTE — FCC 6 GHz Restriction: 47 CFR §15.407(d)(1)(iv) prohibits 6 GHz (5.925–7.125 GHz) access points on aircraft, except that low-power indoor access points and very low power devices may operate in 5.925–6.425 GHz (U-NII-5) in large aircraft while flying above 10,000 feet. MOSAIC 6 GHz radios must therefore be limited to U-NII-5 and must be automatically disabled on the ground and at or below 10,000 feet; the access points or their controller require an altitude or equivalent aircraft input for this purpose. The 2.4 GHz and 5 GHz radios are not affected. Operation outside United States jurisdiction is subject to the rules of the state of registry and the states overflown.

⚠ REFERENCE: RF channel planning and AP placement guidance for the physical cabin environment is addressed in *Passenger Wireless in VIP Aircraft Cabins — Technical White Paper*, Aloft Aeroarchitects, February 2023 (§§4.1–4.3, 6.2).

2.9.11 Network API

Equipment and systems must expose REST API endpoints that are openly shared and documented. All requests must be authenticated.

Method	Purpose	Cabin Example
GET	Retrieve current resource state	Fetch current brightness level of a lighting zone
POST	Create or trigger a resource action	Send a command to adjust cabin temperature

PUT	Update an existing resource	Update IFE software version or scene preset
DELETE	Remove a resource or configuration	Remove a stored user profile from the CMS
Authentication	Secure API access	OAuth 2.0 or API key in Authorization header
Error Handling	Standardized responses	HTTP 4xx/5xx with JSON error body and message

2.9.12 Conformance and Implementation

All components and systems must conform to the defined standards and specifications. Conformance includes regular testing for compliance and continuous updates to address emerging requirements and technologies.

- Pre-installation conformance testing is required for all LRUs.
- Interface conformance test procedures must be published alongside interface specifications.
- Non-conforming devices must not be connected to the MOSAIC network until remediated.

3. Cabin Lighting Systems

This section defines MOSAIC-compliant lighting control for all cabin zones. All lighting, regardless of fixture type or zone, is controlled exclusively by one of two methods: Physical Discrete (direct wired signal) or LAN API (REST or MQTT over the wired cabin network). No DMX, native PWM, Wi-Fi, or proprietary analog protocols are permitted as primary control interfaces.

3.1 Lighting Zone Overview

IP Address Range	Zone	Primary Control	Notes
10.33.YY.ZZZ	Area Lighting	Physical Discrete or LAN API (MQTT / REST)	Zone-level on/off and brightness via API
10.33.10.1–254	Wash Lighting	LAN API (MQTT / REST)	Zone-level dimming via API command
10.33.YY.ZZZ	Table and Artwork	Physical Discrete or LAN API (MQTT / REST)	Individual or zone control
10.33.YY.ZZZ	Reading Lights	Physical Discrete or LAN API (MQTT / REST)	Per-seat discrete on/off; network required for RGB
10.33.YY.ZZZ	Toe Kick	LAN API (MQTT / REST)	RGB color and brightness via WLED API
10.33.YY.ZZZ	Task Lighting	LAN API (MQTT / REST) or Discrete	Discrete for simple on/off; API for dimming
N/A	Closet and Storage	Physical Discrete	Door-switch triggered; no network control required
10.33.YY.ZZZ	Effects / RGB Accent	LAN API (MQTT / REST)	WLED firmware; full color, effect, and preset control

NOTE: The third octet (YY) of each address above is the ATA sub-chapter of the lighting subsystem per §2.9.2; YY codes other than 33-10 (wash lighting) are pending assignment (§2.9.6). YY blocks do not create sub-subnets — all addresses remain on VLAN 33 / 10.33.0.0/16.

NOTE — Extended IP Range (>254 devices in one subsystem): When a subsystem's block (10.33.YY.1–10.33.YY.254) is exhausted, continue in the next consecutive block (e.g., 10.33.11.1 for wash lighting). The /16 subnet (10.33.0.0/16) provides 65,534 host addresses across 256 /24 blocks, all of which are on VLAN 33 and route identically — no sub-netting, VLAN change, or switch reconfiguration is required. Sub-chapter code assignments must leave the following block unassigned for any subsystem that may exceed 254 devices. See Figure 2.9.9 for the multi-switch topology that supports this addressing scale.

3.2 Control Interface Requirements

Each lighting controller must implement at least one of the following interfaces. Where both are implemented, LAN API takes precedence for all commanded state changes; physical discrete is retained as a fallback for power-on defaults and emergency override.

Interface	Requirements
Physical Discrete	Active high (>22 VDC preferred) or active low (ground). On/off only. No brightness or color control. Minimum requirement for all lighting zones.
LAN API — MQTT	Connect to VLAN 33 / 10.33.0.0/16 via wired Ethernet (§2.9.9). Publish state; subscribe to command topics. Topic naming per §2.2.5.
LAN API — REST	Expose HTTP endpoints on port 80 (or 443 with TLS). Respond to GET (state), POST (command). Comply with §2.9.11 API standards.

3.3 Cabin Lighting Modes

Where MQTT is used, all wired network-connected lighting zones must respond to cabin mode commands published by the CMS on the MQTT topic:

```
mosaic/44/cms/cabin-mode/set
```

Cabin Mode	Lighting Behavior
BOARDING	Area lighting at 100%, warm white. RGB effects off or subtle.
TAXI / TAKEOFF	Area lighting at 50%. RGB effects off. Reading lights off.
CRUISE	Area lighting at configurable level per zone. RGB effects enabled.
DINING	Warm dimmed ambient. Task lighting at 80%. RGB effects subtle.
SLEEP	Area lighting off or night-light level. Toe kick at <5%.
LANDING	Area lighting at 100%. RGB effects off.
EMERGENCY	Full white at 100% on all wired LAN zones. Discrete-controlled zones switch on via hardwired signal. Overrides all other commands.

NOTE: Emergency mode must be activatable via physical discrete signal, if available, independent of network availability. Every zone with discrete capability must wire the EMERGENCY line to a dedicated discrete input that bypasses LAN command state. Emergency mode is not meant to replace FAA or EASA compliance with cabin emergency lighting regulations.

3.4 WLED-Based RGB Lighting Control

MOSAIC-compliant lighting controllers for all RGB accent, effects, and toe kick zones (ATA 33) must run WLED open-source firmware, providing HTTP REST and MQTT interfaces directly compatible with the MOSAIC LAN API. Controllers may be single-channel or multi-channel depending on zone requirements. All controllers must accept aircraft bus power (28 VDC or 115 VAC, 360–800 Hz wild frequency as applicable) and connect to VLAN 33 via wired Ethernet. WLED-based controllers meeting the environmental, electrical, and interface requirements of this specification may be used, subject to engineering approval. Refer to the specific product data sheet for connector types, pin assignments, and mating hardware. Although some controllers support sACN as a hardware-level input protocol, sACN must not be used as the primary MOSAIC control interface. Per §3.0, all commanded state changes must use the LAN API (MQTT or REST). The sACN input must be left unconfigured or disabled in WLED settings on all MOSAIC deployments. Figure 3.4.1 (MQTT Message Flow) illustrates how commands and health topics flow between the CMS, MQTT broker, lighting controllers, and the Master Server health agent. The full topic namespace is described in §2.2.5.

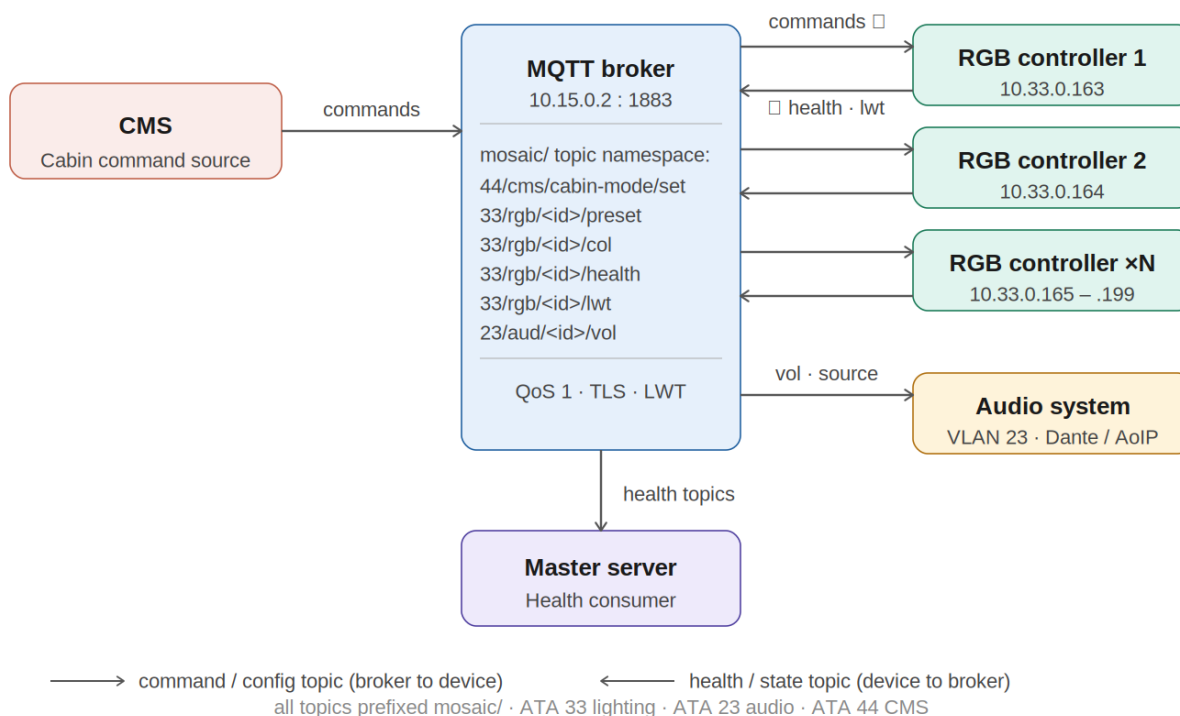


Figure 3.4.1 — MOSAIC MQTT Message Flow

3.4.1 Network Integration

Each controller is assigned a static DHCP lease at its 10.33.YY.ZZZ address (§2.9.2) within VLAN 33 / 10.33.0.0/16 via MAC address reservation (§2.9.7) and registers with the MQTT broker on the Master Server at startup. Power the controller from the aircraft bus supply (28 VDC or 115 VAC wild frequency) as appropriate for the installation. Connect the Ethernet port to a VLAN 33 access switch port. Strip power outputs supply 24 VDC to LED strips. Refer to the controller product data sheet for connector part numbers, pin assignments, and mating hardware.

MQTT Topics (per controller)

Topic	Direction	Description
<code>mosaic/33/rgb/<id>/state</code>	Subscribe	Full JSON state payload (on, brightness, color, effect)
<code>mosaic/33/rgb/<id>/command</code>	Publish	Send full or partial JSON state update
<code>mosaic/33/rgb/<id>/brightness</code>	Publish	Set brightness 0–255
<code>mosaic/33/rgb/<id>/color</code>	Publish	Set RGB hex string e.g. #FF8C00
<code>mosaic/33/rgb/<id>/effect</code>	Publish	Set effect by name or ID
<code>mosaic/33/rgb/<id>/preset</code>	Publish	Recall saved scene preset by integer ID

mosaic/33/rgb/<id>/lwt	Subscribe	Last Will — controller offline indicator (§2.2.5)
------------------------	-----------	---

3.4.2 WLED REST API

WLED exposes a REST API on port 80 compliant with §2.9.11. Key endpoints:

Method	Endpoint	Description	
GET	http://<ip>/json/state	Returns full JSON state: on, brightness, color, effect, segments	
GET	http://<ip>/json/info	Returns device info: firmware version, LED count	
POST	http://<ip>/json/state	Send partial or full JSON state update to controller	
GET	http://<ip>/json/effects	Returns list of all available effects by name and ID	
GET	http://<ip>/json/palettes	Returns list of available color palettes	
Method	Endpoint	Category	Description
GET	/json	State + Info	Full combined response: state, info, effects list, palettes list. Use for initial device poll.
GET	/json/state	State	Current light state: on/off, brightness (bri), color (seg[].col), effect (seg[].fx), palette (seg[].pal), segments.
POST	/json/state	Control	Set light state. Send partial or full JSON body. Returns updated state. Primary control path for MOSAIC.
GET	/json/info	Health/Info	Device info: firmware version (ver), uptime (uptime), free heap (freeheap), Ethernet link state (eth_link), LED count (leds.count), temperature if sensor present.
GET	/json/effects	Discovery	Returns JSON array of all effect names indexed by effect ID. Cache this list; it changes only on firmware update.

GET	/json/palettes	Discovery	Returns JSON array of all palette names indexed by palette ID.
GET	/json/live	Diagnostics	Real-time LED color data stream (n, leds array). Used for visual verification and diagnostics. High-frequency polling not recommended.
GET	/json/nodes	Discovery	Returns discovered WLED nodes on the same network segment. Useful for Master Server auto-discovery of all RGB controllers.
GET	/win	Legacy/Simple	HTTP API shorthand. Accepts query parameters: T (toggle), A (brightness 0–255), R/G/B (color channels), FX (effect ID), FP (palette ID), PL (preset ID). Example: /win&T=1&A=200
GET	/win&T=1	Control	Toggle on/off (T=1 toggle, T=0 off, T=2 on).
GET	/win&PL=<n>	Presets	Recall preset number n. Maps directly to cabin mode presets (§3.4.4). Fastest path for CMS scene recall over REST.
POST	/json/state (preset)	Presets	Recall preset via JSON: {"ps":1}. Equivalent to /win&PL=1. Preferred method for MOSAIC REST preset activation.
GET	/settings	Config	Settings page (HTML). Not used programmatically; for maintenance browser access only.
GET	/settings/ethernet	Config	Wi-Fi settings form (HTML). Use during commissioning to confirm SSID and IP assignment.
GET	/update	Maintenance	OTA firmware update page. Access restricted to maintenance windows (§3.4.5). Master Server automates updates via HTTP POST to this

			endpoint with firmware binary.
POST	/upload	Maintenance	Upload preset or configuration file (JSON). Used by Master Server to restore backup presets to replacement units (§3.4.4).

NOTE: WLED's MQTT topic prefix is configurable. Set the MOSAIC topic root (mosaic/33/rgb/<id>) in WLED Settings > Sync > MQTT so all native WLED publish/subscribe topics align with the MOSAIC naming convention. The <id> field must match the controller's assigned last-octet IP suffix.

3.4.2a Standard WLED Commands — REST HTTP API

The following table covers the complete set of standard WLED HTTP endpoints. All endpoints operate on port 80 (HTTP) or 443 (HTTPS if TLS is enabled). The JSON API is the primary control and status path; the win (HTTP API) path is a legacy shorthand for simple control.

Method

3.4.2b WLED MQTT Command Reference

WLED subscribes and publishes to the following MQTT topics. Configure the topic prefix in WLED settings to mosaic/33/rgb/<id> to align with the MOSAIC namespace. The subtopics below are appended to that prefix automatically by WLED firmware.

Topic (relative to prefix)	Direction	Values	Description
/v	Publish	0–255	WLED publishes current brightness on every state change. Master Server subscribes to detect any lighting change.
/c	Publish	Hex string	WLED publishes current primary color (e.g. #FF8C00) on state change.
(root)	Subscribe	ON / OFF / T / 0–255	Send ON, OFF, T (toggle), or a brightness integer 0–255 to control the light. Simplest control path.
/col	Subscribe	Hex string	Set primary color. Example: #FF0000 for full red.
/api	Subscribe	JSON or win params	Pass any WLED HTTP API payload via MQTT. Enables full JSON state control without HTTP. Example: {"bri":128,"on":true}
/preset	Subscribe	Integer	Recall a stored preset by number. MOSAIC CMS uses this for cabin mode scene changes (§3.4.4).

Example — set zone brightness and color via REST:

```
POST http://10.33.10.163/json/state
{"on":true,"bri":180,"seg":[{"col":[255,140,0],0,0],"fx":0}]}
```

3.4.3 Segments

A single WLED controller may manage multiple independent lighting sub-zones on one LED string using segments. Each segment maps to a contiguous range of LEDs and can be set to a different color, brightness, and effect independently.

- Segment assignments must be documented in the LRU interface manifest for each aircraft zone.
- Segment 0 is the full-string default. Additional segments are defined by LED index start and end.

3.4.4 Preset and Scene Management

For smaller applications, scenes may be stored on the local device. For new, fully integrated cabins, scenes should be called from the controlling CMS via API (preferred) or by other means.

Where scenes are stored on the local device, the following apply:

- Presets are stored in the WLED controller's onboard flash memory.
- Each cabin mode (§3.3) must have a corresponding preset ID stored on every WLED unit in the aircraft.
- Preset IDs must be standardized across all units — e.g., Preset 1 = BOARDING on every controller.
- The CMS triggers presets by publishing the preset integer ID to `mosaic/33/rgb/<id>/preset`.
- Backup preset data must be maintained on the Master Server and re-applied to replacement units during maintenance.

3.4.5 OTA Firmware Updates

- WLED supports over-the-air firmware updates via HTTP or Arduino OTA protocol.
- The Master Server must host a version-controlled WLED firmware repository.
- Firmware updates must be performed during ground maintenance windows only — never in flight.
- Each controller must report its firmware version after update, via the MQTT health topic where MQTT is used or via the REST API otherwise. Version mismatches trigger a Priority 3 fault (§5.0).

3.4.6 WLED Health Reporting

WLED controllers support health reporting via three complementary transport methods: MQTT publish, REST API polling, and Physical Discrete signal. All three may be implemented simultaneously; the Master Server should consume all available paths.

3.4.6a Health via MQTT (Primary)

WLED publishes state changes automatically to its configured MQTT topic. The Master Server subscribes to the following topics for passive health monitoring:

MQTT Topic	Direction	Health Relevance
mosaic/33/rgb/<id>/v	Publish	Brightness value (0–255). Absence of updates for >90 s indicates controller fault or network loss → P3.
mosaic/33/rgb/<id>/c	Publish	Color change events. Used to verify commanded color was applied.
mosaic/33/rgb/<id>/lwt	Publish	Last Will and Testament payload ("offline"). Broker publishes this automatically if controller disconnects unexpectedly → immediate P3 fault.
mosaic/33/rgb/<id>/health	Publish	MOSAIC health JSON payload (see §5.2). Published at minimum 60-second interval by WLED firmware with MOSAIC extension build, or by a sidecar health agent on the Master Server polling /json/info.

Minimum MOSAIC health payload for WLED controllers:

```
{ "ts": "2026-05-15T14:32:00Z", "status": "ok", "faults": [], "uptime_s": 86400, "fw": "0.15.0", "bri": 180, "eth_link": true, "free_heap": 32768, "leds": 144 }
```

Field	Source	Description
ts	/json/info.time or NTP	UTC timestamp of the health report.
status	Derived	ok, degraded, or fault based on fault array contents.
faults	Derived	Array of active fault objects {"pri":P3,"code":"MQTT_RECONNECT","ts":"..."} . Empty array = healthy.
uptime_s	/json/info.uptime	Seconds since last boot. Unexpected resets trigger a P3 event.
fw	/json/info.ver	WLED firmware version string. Version mismatch vs. Master Server repository → P3 (§3.4.5).
bri	/json/state.bri	Current brightness 0–255. Confirms commanded state was applied.
eth_link	/json/info.eth.link	Ethernet link state (boolean). false = link down → P3 Communication Failure.
free_heap	/json/info.freeheap	Free memory in bytes. Below 10 000 bytes → P4 fault; indicates memory leak or overload.
leds	/json/info.leds.count	Configured LED count. Mismatch vs. manifest → P3 configuration fault.

3.4.6b Health via REST API (Polling)

The Master Server polls each WLED controller's /json/info endpoint at a maximum interval of 60 seconds to verify reachability and extract health fields. Polling supplements MQTT and provides health data when MQTT is unavailable.

Endpoint	Method	Health Fields Extracted
/json/info	GET	ver (firmware), uptime, freeheap, eth_link (Ethernet link state), leds.count, leds.pwr (power draw mA). HTTP timeout or non-200 response → P3.
/json/state	GET	on (power state), bri (brightness), seg[].fx (active effect). State mismatch vs. last command → P4.
/json	GET	Combined state + info in single request. Use for initial poll after commissioning or reconnection.

- HTTP 200 with valid JSON = reachable and operational.
- HTTP timeout (>5 s) or connection refused = P3 Communication Failure.
- HTTP 200 but freeheap < 10 000 = P4 Parameter Bounds. eth_link = false = P3 Communication Failure.
- Firmware version mismatch vs. Master Server repository = P3 (log and schedule OTA update).

3.4.6c Health via Physical Discrete

WLED controllers that support a configurable GPIO output may assert a Fault discrete signal on a dedicated output pin. This provides health visibility independent of the network.

Signal	Logic Level	Meaning
Fault Output (GPIO)	High (>22 VDC) — normal	Controller powered, firmware running, no active P1/P2 faults.
Fault Output (GPIO)	Low (ground) — fault	Controller fault: P1 sub-system failure or P2 power fault. Triggers discrete input on zone monitor or Master Server I/O module.
Power Present	High (>22 VDC) — normal	Controller supply voltage present. Loss of this signal = P2 Power Failure.
Power Present	Low (ground) — fault	Controller unpowered. Logged as P2 immediately.

- The Fault GPIO pin must be configured in WLED's hardware settings and documented in the LRU interface manifest.
- Discrete health signals must be wired to the zone monitor or a dedicated I/O module on VLAN 33 that publishes state to the Master Server via MQTT or another interface defined in §2.2.
- Discrete health is the fallback path when VLAN 33 network connectivity is lost.

5. System Health Monitoring and Reporting

All LRUs and subsystems must report health status to the Master Server. The following priority levels are defined:

Priority	Fault Class	Description
P1	Sub-System Failure	Complete loss of a cabin subsystem (e.g., all lighting in a zone, IFE offline)
P2	Power Failure	Loss of power to one or more LRUs or distribution segments
P3	Communication Failure	LRU MQTT disconnect, LWT received, or network timeout
P4	Parameter Bounds	Sensor reading or operational parameter outside defined limits

5.1 Sub-System Logging

Each LRU must log and report the following minimum data to the Master Server:

- Timestamp (UTC, NTP-synchronized)
- Power-on count
- Runtime hours
- Fault event log (priority, code, timestamp)
- Firmware version
- Network address and interface status

5.2 Health Reporting — MQTT (Where Implemented)

LRUs that implement MQTT must publish health status on the following topic at a minimum interval of 60 seconds:

```
mosaic/<ata>/<subsystem>/<lru-id>/health
```

Minimum JSON payload:

```
{ "ts": "2026-05-15T14:32:00Z", "status": "ok", "faults": [], "uptime_s": 86400, "fw": "0.15.0", "bri": 180, "eth_link": true, "free_heap": 32768 }
```

- The Master Server aggregates all health payloads and produces a cabin-level health summary.
- P1 and P2 faults must trigger an immediate retained publish on the health topic and an alert on the crew interface.
- All health data must be stored in the system event log with a minimum retention of 1000 flight cycles or 3 years, whichever is greater.

5.3 Health Reporting — REST API Polling

The Master Server polls each LRU's REST API at a maximum interval of 60 seconds as a health check. REST polling provides health data when MQTT is unavailable or not implemented and is mandatory for all LRUs that expose an HTTP interface.

Endpoint	Method	LRU Type	Purpose
/json/info	GET	WLED RGB	Firmware version, uptime, free heap, Ethernet link state (eth_link), LED count. Full WLED health fields (§3.4.6b).
/json/state	GET	WLED RGB	Current on/off, brightness, active effect/palette, segment state. Confirms commanded state was applied.
/api/v1/status	GET	General LRU	MOSAIC standard status endpoint (§5.5). Returns LRU identity, power state, fault list, uptime. See Appendix B.
/api/v1/health	GET	General LRU	Lightweight reachability ping. Returns {"status":"ok"} with HTTP 200 when operational. HTTP timeout = P3.
/api/v1/faults	GET	General LRU	Active fault log. Returns array of current faults with priority, code, and timestamp.

- A non-200 HTTP response or TCP connection timeout (>5 s) from any LRU is logged as a P3 Communication Failure.
- Where MQTT is implemented, REST polling supplements it: MQTT provides real-time event-driven updates and REST provides scheduled verification. Where MQTT is not implemented, REST polling is the primary health reporting method.
- The Master Server must stagger polling across all LRUs to avoid simultaneous burst traffic on VLAN 33.

5.4 Health Reporting — Physical Discrete

Physical discrete signals provide health monitoring that is independent of the cabin network. All LRUs with discrete interface capability must implement at minimum a Power Present output. LRUs with processing capability should also implement a Fault output.

Signal Name	Logic (Active High)	Fault Priority	Description
Power Present	High = power OK; Low = no power	P2	DC supply voltage present at the LRU input. Loss triggers a P2 Power Failure immediately. All LRUs must implement this signal.
Fault / Ready	High = operational; Low = fault	P1	General fault indicator. Low state indicates an active P1 Sub-System Failure. Must be wired to zone monitor or dedicated I/O.

Network Ready	High = network OK; Low = no network	P3	Optional. Indicates LRU has an active IP network connection. Use where MQTT LWT alone is insufficient (e.g. Ethernet-only devices).
Override Active	High = override; Low = normal	Info	Optional. Indicates emergency override is active on this LRU. Published to MQTT as an informational event.

- Active high logic (>22 VDC = healthy) is preferred per §2.2.3.
- Discrete signals must be wired to a zone monitor or I/O module that publishes state changes to `mosaic/<ata>/<subsystem>/<lru-id>/discrete` via MQTT, or reports them through another interface defined in §2.2.
- The Master Server must correlate discrete fault events with MQTT health payloads, where present, to avoid duplicate P-level logging.
- For lighting zones that use Physical Discrete as their sole control method (§3.2), the Power Present discrete is the only required health signal.

5.5 Alerts and Notifications

⚠ REVIEW REQUIRED: Needs definition based on cabin server set up.

6. Audio Integration

This section defines MOSAIC-compliant audio distribution for the aircraft cabin. Two methods are defined: a basic digital audio path used where network audio infrastructure is unavailable, and Audio over IP (AoIP), the preferred method for all new installations. Both methods integrate with the MOSAIC MQTT control fabric for zone volume and source control.

6.1 Audio Methods Summary

Method	Architecture	Preferred Use
Basic Digital Audio	HDMI source → AV matrix switch or node (audio extraction) → digital amplifier → speaker	Retrofit, low-channel-count zones, or where network audio infrastructure is not available. Minimum acceptable implementation.
AoIP / AES67 (preferred)	Audio source → AES67-compliant audio switch → network amplifier / powered speaker	All new installations. Supports flexible zone routing, low latency, dynamic re-patching without rewiring, and MQTT-driven volume and source control.

6.2 Basic Digital Audio (Minimum Implementation)

The basic digital audio path is the fallback implementation where AoIP infrastructure is not available. It uses HDMI audio extraction at the AV matrix switch or node to feed the cabin amplifiers.

This section assumes that all audio sources originate from digital media. Purely analog sources, such as RCA line-level sources or similar, are not explicitly covered.

6.2.1 Signal Chain

The typical signal chain is shown in Figure 6.2.1. An HDMI source, such as a Roku or Apple TV media player or an IFE server, feeds the system. Audio is extracted at the AV matrix switch or node and is then distributed over the cabin network as AoIP (§6.3) or to a digital amplifier that feeds the cabin speakers.

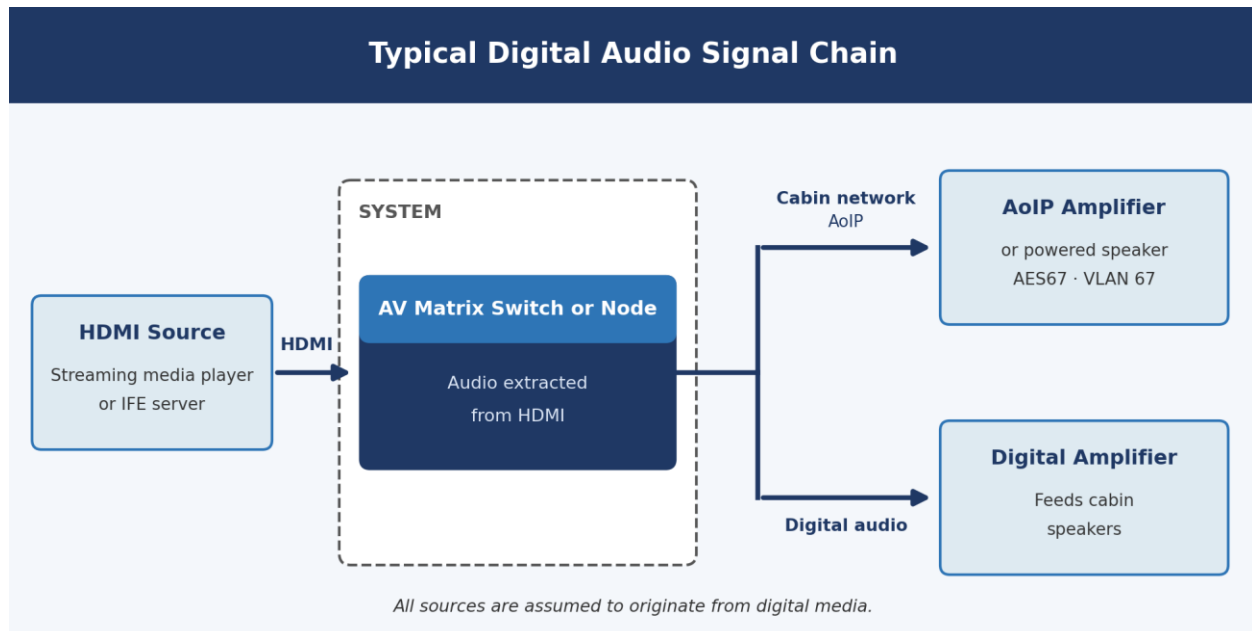


Figure 6.2.1 — Typical digital audio signal chain. An HDMI source feeds the system, where audio is extracted at the AV matrix switch or node and distributed over the cabin network as AoIP or to a digital amplifier.

NOTE: eARC-capable HDMI switches should be used where available to support multi-channel audio extraction from a single HDMI run. Single-link HDMI extraction yields stereo only.

6.2.2 Limitations

Audio is controlled first by the controlling CMS, which commands the matrix switch or node.

- Analog signal runs are susceptible to EMI/RFI from aircraft systems. Shielded cable (minimum 0.5 mm² twisted pair, shield grounded at one end) is mandatory for all analog runs exceeding 2 m.
- No dynamic MQTT zone control — volume is controlled via analog potentiometer or fixed resistor network only.
- No cabin mode integration — audio mute and source selection must be implemented via physical discrete lines where required.

6.3 Audio over IP / AES67 (Preferred Method)

AES67 is the primary transport protocol for cabin audio over IP. AES67 is an open, standards-based audio-over-IP interoperability standard (RTP/UDP multicast streaming, SAP/SDP session discovery, IEEE 1588-2008 PTP clock synchronization) that avoids dependence on any single vendor's proprietary audio transport. Native AES67 audio hardware is preferred over Dante (by Audinate) hardware. Dante hardware is accepted with the additional network changes of §6.3.6; every Dante device carrying cabin audio must be configured for AES67 mode so that the primary transport between devices is standards-based AES67 rather than native Dante-only flows. AES67 transport enables flexible zone routing, dynamic re-patching from the audio switch or Master Server, and MQTT-driven volume and source control from the CMS.

NOTE: AES67 is preferred over Dante. Dante hardware is accepted only with the additional network changes listed in §6.3.6.

6.3.1 Network Requirements

Requirement	Specification
VLAN	Dedicated audio VLAN (separate from VLAN 33 lighting). Isolates AES67 multicast audio from other cabin traffic. All AES67 and Dante devices are assigned VLAN 67.
IP Subnet	All AES67 and Dante devices are on subnet 10.67.0.0/16 (VLAN 67, gateway 10.67.0.1). DHCP preferred for AES67 devices; static addressing acceptable. All AES67 devices on the same subnet for primary discovery.
Switch	Gigabit Ethernet with IGMP snooping v2 or v3 enabled, per the backbone/distribution switch requirements of §2.9.9. AES67 uses RTP/UDP multicast for audio and SAP/SDP for discovery — IGMP snooping prevents multicast flooding to non-audio ports. QoS-capable switch required (see §6.3.2).
Latency	AES67 supports 0.25 ms, 1 ms, and 5 ms link-offset presets. Use 1 ms as the cabin default; 5 ms is acceptable for background music zones. Sub-millisecond requires careful switch selection — verify with the device manufacturer's qualification tools.
PTP / Clock sync	AES67 requires IEEE 1588-2008 (PTPv2) using the AES67 Media Profile, with a designated PTP grandmaster on the audio VLAN. Backbone and distribution switches carrying the audio VLAN must operate as PTP boundary clocks or end-to-end transparent clocks (see §2.9.9). PTP must be confined to the audio VLAN. Full requirements are given in §6.3.5.
Redundancy	Qualifying devices provide a redundant network port. Connect the primary port to the primary audio switch and the secondary port to a separate switch or path where zero-downtime is required.

△ NOTE — Multicast Flooding: AES67 and Dante devices send audio, PTP clock, and discovery traffic as multicast, and typically ship with DHCP addressing and PTP domain 0 enabled, so they begin transmitting as soon as they are connected. If the switch port for such a device is not assigned to VLAN 67 — for example, it is left on the switch default VLAN or assigned to another subsystem VLAN — or if IGMP snooping is not active on VLAN 67, this multicast traffic is flooded to every port in the VLAN it lands in and can degrade or disrupt other cabin subsystems. Every AES67 or Dante device port must be configured as a VLAN 67 access port before the device is connected (§2.9.2), every trunk carrying VLAN 67 must tag it, and IGMP snooping (v2 or v3) with an active querier and a short query interval must be enabled on VLAN 67 on every switch in the audio path.

6.3.2 QoS Configuration

AES67 audio traffic must be prioritized over best-effort data on the audio VLAN. Configure the following DSCP markings on the audio switch:

Traffic Type	DSCP Marking	802.1p Priority / Notes
AES67 audio (unicast)	AF41 (34)	Priority 6. Second-highest queue, below PTP. Strict priority scheduling.

AES67 audio (multicast)	AF41 (34)	Priority 6. Ensure multicast queues are not starved.
PTP clock messages	EF (46)	Priority 7. Highest queue. Clock sync messages must not be delayed.
MQTT / management	CS1 (8)	Best-effort or low priority on audio VLAN.

NOTE: AES67 audio frames are 100-byte UDP packets at 1 ms intervals (1 kHz packet rate per channel). Jitter above ~100 μ s causes audible glitches. Strict priority queueing (SPQ) on the audio switch is strongly preferred over weighted fair queueing (WFQ) to guarantee latency bounds.

NOTE: The markings above are the AES67 standard defaults. Dante devices mark traffic differently; see §6.3.6.

6.3.3 Device Types

Device Type	Role
AES67 audio switch	Routes AES67 audio streams between transmitters and receivers. May also perform analog I/O conversion. Acts as the central hub for all cabin audio routing. Receives MQTT vol/source commands from the Master Server.
IFE / media server	AES67 transmitter. Sends audio channels (stereo music, PA announcement, IFE audio per seat zone) over the audio VLAN. Connects to audio switch via VLAN-tagged Ethernet port.
Network amplifier / powered speaker	AES67 receiver. Receives audio from the switch, amplifies, and drives the cabin speaker. Eliminates the separate analog amplifier and long analog cable runs of the basic path.
Software audio endpoint	Software AES67 endpoint (Dante Virtual Soundcard where Dante hardware is used) running on the Master Server for monitoring, tone injection, or announcement playback.
Dante Controller (Dante hardware only)	PC application for commissioning Dante devices and confirming AES67 mode on each device. Used on the maintenance laptop to patch transmitter channels to receiver channels. Routing is stored in device flash and survives power cycle. Not required in-service.

6.3.4 Audio Zone Topology

Each cabin audio zone corresponds to one or more AES67 receiver channels on the network amplifier or powered speakers for that zone. Zone assignments are patched in the device configuration tool (Dante Controller where Dante hardware is used) during commissioning and stored in device memory. The CMS may re-route zones in service via MQTT commands to the audio switch MQTT interface (see §6.4).

- Typical zones: forward cabin music, aft cabin music, galley PA, lavatory chime, crew interphone.
- AES67 audio mutes by default during a PA announcement. AES67 audio is not required to carry or override for PA, on the assumption that a certified PA system is incorporated.
- AES67 audio is not intended for certified passenger address at this time (see §6.5).

6.3.5 PTP Clock Synchronization Requirements

The following requirements refine the PTP / clock sync requirement of §6.3.1 and apply to every AES67 device and every switch carrying the audio VLAN.

Parameter	Requirement
Standard and Profile	IEEE 1588-2008 (PTPv2) using the AES67 Media Profile (AES67 Annex A). PTPv2 is the reference clock for all AES67 streams.
Transport	UDP over IPv4 multicast, group address 224.0.1.129, UDP port 319 (event messages) and UDP port 320 (general messages).
Domain	A single PTP domain on the audio VLAN. Domain number 0 (AES67 default) unless otherwise specified per program. All AES67 devices and PTP-aware switches on the audio VLAN must use the same domain number.
Delay Mechanism	End-to-end (E2E) delay request-response on all devices and switches. Peer-to-peer (P2P) delay must not be used; E2E and P2P devices do not interoperate within one domain.
Message Intervals	AES67 Media Profile defaults: Sync logSyncInterval = -3 (125 ms); Announce logAnnounceInterval = 1 (2 s); announceReceiptTimeout = 3; Delay_Req logMinDelayReqInterval = 0 (1 s). All devices in the domain must use identical Announce interval and timeout values.
Grandmaster	One designated primary grandmaster on the audio VLAN; a designated backup grandmaster is preferred. The primary must hold the lowest priority1 value in the domain and the backup the next lowest, so that the Best Master Clock Algorithm (BMCA) elects them in that order. All other devices must remain at the profile default priority1 (128) or higher, or be configured slave-only where supported.
Switch Clock Mode	Every switch in the path between the grandmaster and any AES67 device must operate as a boundary clock (preferred) or E2E transparent clock with hardware timestamping on all ports carrying the audio VLAN (§2.9.9). Switches without PTP support are not permitted in the audio path.
VLAN Confinement	PTP messages must not leave the audio VLAN (§2.9.2). PTP must be disabled on switch ports that do not carry the audio VLAN, and the Layer 3 control point (§2.9.2) must not forward PTP between VLANs. Any other VLAN that later requires PTP must use a separate PTP instance with its own domain number and grandmaster.
Accuracy	Each AES67 device must hold its clock offset from the grandmaster within $\pm 1 \mu\text{s}$ in steady state.
QoS	PTP event messages must be marked per §6.3.2 and served from a strict-priority queue on every switch in the audio path.
Monitoring	Each AES67 device reports clock lock status via MQTT (§6.4).

⚠ REVIEW REQUIRED: The grandmaster device(s) and their priority1/priority2 values have not been selected for this program, and the $\pm 1 \mu\text{s}$ offset limit has not been validated on MOSAIC switch hardware. Both must be confirmed by bench test before delivery.

6.3.6 Dante Acceptance — Additional Network Changes

Native AES67 hardware is preferred. Dante hardware may be used in place of native AES67 hardware provided every Dante device operates in AES67 mode (§6.3) and the following network changes are made:

- PTPv1 coexistence: Dante devices continue to exchange IEEE 1588-2002 (PTPv1) messages among themselves in addition to PTPv2. Switches on the audio VLAN must forward PTPv1 messages unmodified, and PTPv1 remains confined to the audio VLAN per §6.3.5.
- Grandmaster priority: Dante devices take part in the PTPv2 Best Master Clock Algorithm. The designated grandmaster of §6.3.5 must hold a lower priority¹ value than every Dante device so that a Dante device is not elected.
- QoS: Dante devices may mark PTP as CS7 (56) and audio as EF (46) rather than the AES67 markings of §6.3.2. Switch queue mapping must place both sets of markings in the correct strict-priority queues.
- Multicast addressing: Dante AES67 flows use a configurable multicast prefix (239.69.0.0/16 by default). The program multicast address plan and IGMP snooping configuration must cover this range.
- Energy Efficient Ethernet (IEEE 802.3az) must be disabled on every switch port in the audio path.
- Configuration tooling: Dante Controller is required on the maintenance laptop for device configuration and flow setup, and it must connect on the audio VLAN.

6.4 MQTT Integration for Audio Control

Audio system components on the audio VLAN bridge to the MOSAIC MQTT broker via the Master Server. The Master Server translates MQTT commands into AES67 stream/session management calls (or vendor API calls, e.g. Dante, when operating in AES67 mode) or OSC messages to the audio switch.

MQTT Topic	Direction	Function
mosaic/23/aud/<id>/vol	Subscribe (device)	Set zone volume. Payload: integer 0–100. Maps to dB attenuation on the AES67 receiver channel for that zone.
mosaic/23/aud/<id>/src	Subscribe (device)	Set zone audio source. Payload: source identifier string (e.g. "IFE", "PA", "MUSIC"). Triggers AES67 routing re-patch via Master Server bridge.
mosaic/23/aud/<id>/mute	Subscribe (device)	Mute or unmute the zone. Payload: "true" / "false". Used by CMS for cabin-mode state changes (e.g. mute all during PA).
mosaic/23/aud/<id>/state	Publish (device)	Zone state report. JSON object containing current volume, source, mute state, and AES67/PTP clock lock status. Published at 60 s interval and on any state change.
mosaic/44/cms/cabin-mode/set	Subscribe (audio bridge)	Audio bridge subscribes to cabin mode. Maps mode to audio preset: BOARDING → low background music; CRUISE → full IFE; SLEEP → mute or low; EMERGENCY → mute all, enable PA discrete.

NOTE: The <id> field in audio MQTT topics follows the same naming convention as lighting (§2.2.5): mosaic/<ata>/<subsystem>/<zone-id>/<function>. Audio uses ATA 23 and subsystem "aud". Zone IDs must be registered in the Master Server commissioning database.

6.5 Emergency Audio Fallback

NOTE: AES67 audio is not intended for certified passenger address at this time.

6.6 Commissioning

- Install the device configuration tool on the maintenance laptop (Dante Controller where Dante hardware is used). Verify all devices are discovered and that every Dante device is configured for AES67 mode.
- Patch transmitter channels to receiver channels per the zone audio map. Save the routing configuration to all devices.
- Verify PTP/AES67 clock lock on all devices (shown as a green clock icon in Dante Controller where Dante hardware is used).
- Verify PTP per §6.3.5: the designated grandmaster is elected, all AES67 devices report the same grandmaster identity, clock offset is within limit, and no PTP messages are present on any VLAN other than the audio VLAN.
- Configure MQTT topic mappings in the Master Server audio bridge configuration file. Test each zone with a vol command from the commissioning laptop.
- Verify that AES67 audio mutes on all zones during a test PA announcement.

⚠ REVIEW REQUIRED: All AES67 and Dante devices are assigned VLAN 67 / 10.67.0.0/16. The MQTT bridge software has not been specified for this program.

7. Video Distribution

This section defines MOSAIC-compliant video distribution from cabin video sources (IFE, moving map, cameras, and media players) to cabin monitors. The first priority is video transported over fiber; SDI and copper Ethernet are the alternatives at the monitor.

7.1 Priority 1 — Video over IP on Fiber

Requirement	Specification
Target Format	3840 × 2160 at 60 frames per second, 4:4:4 chroma, HDR (10-bit or greater).
Transport Standard	IPMX (VSF TR-10) is preferred. SMPTE ST 2110 is accepted. IPMX is built on the ST 2110 transport, so both use the same RTP stream formats.
Compression	Compression is assumed. JPEG XS (ISO/IEC 21122) carried per VSF TR-10-7 / SMPTE ST 2110-22. Uncompressed transport is not required (see §7.2).
Physical Medium	Fiber.

This specification states the video format requirement only; it does not prescribe how the video source is connected. In current installations the fiber video feed originates from an AV matrix switch, not directly from the 10G Ethernet switch. A future state may connect video directly from the Ethernet switch.

7.2 Bandwidth and Compression

Uncompressed 4K60 4:4:4 video requires approximately 18 Gbps per stream. HDMI Licensing describes the HDMI 2.0 Specification as providing “support of 4K@60fps and 18Gbps bandwidth” (HDMI Licensing, LLC, HDMI 2.0 press release, <https://www.hdmi.org/press/bodydetails/20>). At that link rate 4K60 4:4:4 is carried at 8 bits per component; 4:4:4 with 10-bit HDR requires more.

Compression is therefore assumed for 4K60 4:4:4 HDR transport (§7.1).

7.3 Monitor Interface Priority

The video interface at each monitor follows the order below, with fiber as the default. The fiber connection and monitor interface are provided by the AVDS Node hardware.

Priority	Monitor Interface	Notes
1 (default)	Fiber	Default monitor interface.
2	SDI	12G-SDI (SMPTE ST 2082) over coaxial cable. A single 12G-SDI link carries 2160p60 at 4:2:2 10-bit, not 4:4:4.
2	Copper Ethernet	Alternative to fiber.

8. Component Identification and Labeling

All MOSAIC network components must be permanently and legibly labeled before installation. Labels must contain sufficient information for a technician to identify, locate, and replace the component with a correctly configured unit without reference to external documentation at the point of installation.

8.1 Applicable Standards

Labeling should be robust and clear to the end user. Operation details should be exposed to the installer as much as possible. Reference the following standards:

Standard	Description
MIL-STD-130N	Identification Marking of U.S. Military Property. Governs marking methods, data content, Unique Item Identification (UID/IUID), and machine-readable information (MRI) using 2D Data Matrix ECC 200 barcodes. All MOSAIC LRUs are subject to MIL-STD-130N marking requirements.
MIL-DTL-15024G	Plates, Tags, and Bands for Identification of Equipment. Defines physical label types (Type G metal foil, Type H photosensitive anodized aluminum, Type L polyester), material properties, adhesive requirements, and environmental survivability. Type H (Metalphoto®) or Type L polyester is the preferred medium for MOSAIC cabin installations.
MIL-STD-1472G	Human Engineering. Establishes legibility requirements including minimum character height (2.3 mm / 0.09 in at normal viewing distance), contrast ratio, font selection, and spacing. All label text must comply with §5.4 of MIL-STD-1472G.
MIL-STD-810H	Environmental Engineering Laboratory Tests. Label materials must survive the aircraft cabin environment per: Method 501.7 (High Temperature), Method 502.7 (Low Temperature), Method 507.6 (Humidity), Method 514.8 (Vibration), and Method 504.3 (Contamination by Fluids). Material selection per MIL-DTL-15024G must demonstrate compliance.
MIL-STD-129R	Military Marking for Shipment and Storage. Applies to packaging and shipping labels for LRU spares and replacement units. Governs barcode formats used on outer packaging to correlate shipment contents with installed unit records on the Master Server.

8.2 General Label Requirements

8.2.1 Material

Label material must be selected to survive the full service life of the component in the aircraft cabin environment without delamination, fading, or illegibility.

Material Type	MIL-DTL-15024G Type	Recommended Application
Photosensitive anodized aluminum (Metalphoto®)	Type H	Preferred for all fixed-mount LRUs (controllers, switches, Master Server). Resists fluids, abrasion, and UV. Rated for full aviation service life.

Adhesive-backed metal foil	Type G	Acceptable alternative to Type H on metal chassis. Anodized aluminum or stainless steel foil with permanent pressure-sensitive adhesive.
Polyester film	Type L, Style 1–4	Acceptable for cables, patch panels, and temporary service labels. Must be rated for the applicable temperature range and fluid exposure class per MIL-DTL-15024G.

- Adhesives must be permanent pressure-sensitive and fungus-resistant per MIL-DTL-15024G §3.4.
- Flammable label materials are prohibited per MIL-DTL-15024G §3.3 and FAR Part 25.853.
- Labels must not be painted over, covered, or obstructed after installation.

8.2.2 Format and Legibility

- Minimum character height: 2.3 mm (0.09 in) for labels read at normal arm's length per MIL-STD-1472G §5.4.
- Font: sans-serif (Arial or Helvetica equivalent). No decorative or serif fonts.
- Text color: black on white or silver background. Minimum contrast ratio 3:1 per MIL-STD-1472G.
- All primary identification fields (LRU ID, P/N, S/N) must be uppercase. Mixed case is acceptable for descriptive fields.
- Labels must be printed or engraved. Handwritten labels are not permitted.

8.3 Required Data Fields — All Network Components

Every label must include all mandatory fields. Conditional fields are required where applicable to the component type.

Field	Required	Content and Source
LRU ID / Hostname	RECOMMENDED	Human-readable identifier matching the DHCP hostname, MQTT <lru-id>, and commissioning database. Example: WLED-RGB-FWD-COVE
ATA Chapter / VLAN	MANDATORY	ATA chapter number and VLAN ID. For all lighting: ATA 33 / VLAN 33.
Zone	RECOMMENDED	Lighting zone or subsystem label per §3.1. Example: FWD-COVE, GALLEY-WASH, MAIN-EFFECTS
IP Address	MANDATORY	Static IP assigned per §2.9.7 and DHCP reservation. Example: 10.33.10.163
MAC Address	MANDATORY	Primary Ethernet interface MAC address used for DHCP reservation validation. Example: AA:BB:CC:11:22:33
Manufacturer	RECOMMENDED	Component manufacturer name, used to identify compatible replacement units.
Part Number (P/N)	MANDATORY	Manufacturer part number sufficient to order a direct replacement. Include revision suffix if hardware variants exist.

Hardware Revision (HW REV)	RECOMMENDED	PCB or hardware revision at installation. Required to confirm replacement unit compatibility. Example: HW REV 3.1
Serial Number (S/N)	MANDATORY	Manufacturer serial number. Forms part of the UII for MIL-STD-130N compliance. Must be globally unique within the part number.
Firmware Version at Install	RECOMMENDED	Firmware version at time of installation. Example: WLED 0.15.0. Replacement units must be updated to match the Master Server repository (§3.4.5) before commissioning.
Date of Installation	RECOMMENDED	ISO 8601 date (YYYY-MM-DD). Used for maintenance scheduling and warranty tracking.
Aircraft Tail / Hull ID	RECOMMENDED	Registration or hull identifier of the aircraft. Prevents cross-installation of spares without reconfiguration.
Port / Connector Reference	CONDITIONAL	Required for switches, patch panels, and multi-port components. Identifies the connected device, VLAN, and cable run ID per §8.4.3.
2D Data Matrix (UII)	RECOMMENDED	MIL-STD-130N Data Matrix ECC 200 barcode encoding the full UII. Must be scannable by the Master Server commissioning agent for automated registration.

8.4 Component-Specific Label Requirements

8.4.1 WLED-Based RGB Controllers (ATA 33 / VLAN 33)

In addition to the common mandatory fields in §8.3, WLED-based lighting controllers must include the following additional fields. These fields apply to any WLED-based controller used on MOSAIC.

Additional Field	Content
Output Channels	Number of active LED output channels. Example: 4-CH
CH1–CH4 LED Type	LED strip part number and type per channel. Example: CH1: 400-0001-0 RGBIC-CCT (SPI) CH2: 400-0001-0 RGBIC-CCT (SPI)
CH1–CH4 Pixel Count	Addressable pixels per channel at installation. Example: CH1: 144px CH2: 60px
CH1–CH4 Color Order	Output data protocol per channel. Example: CH1: SPI CH2: RS-485
Supply Voltage	Aircraft bus input and strip output voltage. Example: INPUT 28 VDC (J2) / STRIP OUTPUT 24 VDC (J5–J8)
MQTT Topic Base	Base MQTT topic path per §2.2.5. Example: mosaic/33/rgb/fwd-cove
Preset Map Reference	Reference to cabin mode preset mapping. Example: PRESETS 1–7 PER §3.4.4 — SEE MASTER SERVER

NOTE: The WLED preset configuration and full LED segment map must be stored in the Master Server commissioning database (§3.4.4) and re-applied to a replacement unit before it is returned to service. The label provides the physical replacement parameters; the Master Server provides the logical configuration. For single-channel controllers, channels CH2–CH4 are N/A; label CH1 only.

8.4.2 Master Server / Router (ATA 15 / VLAN 15)

Additional Field	Content
VLANs Served	All VLAN IDs actively configured on this unit. Example: VLAN 15, 16, 17, 21, 23, 24, 25, 26, 33, 38, 44
Management IP	Management interface IP address. Example: 10.15.0.1
MQTT Broker IP	IP address of the MQTT broker instance. Example: 10.15.0.2
Config Backup Location	Physical location or path of the configuration backup. Example: /etc/mosaic/backup or removable flash card slot label.

8.4.3 Distribution and Access Switches

Each switch chassis label carries the common fields from §8.3. Each populated port must additionally be labeled at the faceplate:

Additional Field	Content
Port Number	Physical port designation matching the switch OS identifier. Example: GE0/1
Connected Device (LRU ID)	LRU ID of the device connected to this port. Example: WLED-RGB-FWD-COVE
VLAN Assignment	Access VLAN (untagged) or trunk VLAN list. Example: VLAN 33 (untagged) or TRUNK: 15,33,44

NOTE: Port labels may be applied as a label strip adjacent to the port row rather than on individual port collars, provided the correspondence between label and port is unambiguous. The strip must be permanently fixed and must not slide or rotate.

8.4.4 Cables and Patch Leads

Every cable must be labeled at both ends. Labels must remain legible and firmly attached through the service life of the cable.

Field	Content
Cable Run ID	Unique identifier cross-referencing both endpoint labels and the cable schedule. Example: CBL-SWA-P01-WLED163
From (Device : Port)	Origin LRU ID and port. Example: SW-A : GE0/1
To (Device : Port)	Destination LRU ID and port. Example: WLED-RGB-FWD-COVE : ETH0
Cable Type	Physical medium. Example: CAT6A, SPE 10BASE-T1L, SFP+ OM3 FIBER
VLAN	VLAN carried on this cable. Example: VLAN 33
Length	Installed cable length in metres. Example: 4.5 m

8.4.5 Power Supplies and DC Distribution

Additional Field	Content
Input Voltage / Current	Nominal aircraft bus input. Example: 28 VDC / 5 A
Output Voltage / Current	Regulated output to the LRU. Example: 12 VDC / 10 A
Circuit Breaker / Fuse ID	Upstream protective device identifier. Example: CB-LIGHTING-FWD-12V
Connected LRU(s)	LRU IDs powered by this supply. Example: WLED-RGB-FWD-COVE, WLED-RGB-MID-COVE

8.5 Label Placement

- Labels must be affixed to the component body on a surface visible when the unit is installed in its normal service position, without requiring removal of adjacent equipment.
- Where the primary label is obscured by mounting hardware or adjacent structure, a secondary abbreviated label (LRU ID, IP address, and MAC address minimum) must be placed on a visible surface.
- Labels must not be placed over ventilation apertures, connector seals, or safety markings.
- Cable labels must be applied within 50 mm of each connector or termination point. Heat-shrink sleeves over the label are permitted provided they do not reduce legibility.
- Switch port labels must be readable with the cable installed in the port.

8.6 Standard Label Template

The following layout is the minimum required content arrangement for a WLED controller. The Data Matrix barcode occupies the lower-right area. For small components where the full template does not fit, the minimum acceptable label is: LRU-ID, IP, MAC, S/N, and Data Matrix barcode.

```

+-----+-----+-----+
| LRU-ID:  WLED-RGB-FWD-COVE |      |      |
| P/N:      [CTRL-P/N-HERE]  |      |      |
| HW REV:   DRAFT  S/N: SN000001 | [DATA  |
| FW:       WLED 0.15.0       | MATRIX |
| INST:     2026-05-01       | UII]  |
| IP:       10.33.10.163      |      |
| MAC:      AA:BB:CC:11:22:33 |      |
+-----+-----+-----+
| ATA 33 / VLAN 33 / FWD-COVE | AIRCRAFT |
| MFR: [Manufacturer Name]    | G-ABCD  |
+-----+-----+-----+
| CH1: RGBIC-CCT SPI 24VDC-OUT MQTT:      |
| CH2: RGBIC-CCT SPI 28VDC-IN  mosaic/33/ |
| CH3: --  CH4: --             fwd-cove   |
+-----+-----+-----+

```

8.7 Commissioning and Replacement Procedure

The label is the primary physical record of a unit's identity and configuration. The following procedure must be followed when installing a new or replacement unit:

- Scan the Data Matrix barcode on the replacement unit to confirm the P/N and HW REV match the unit being replaced. Do not install a unit with a different HW REV without engineering approval.

- Verify the MAC address on the replacement unit label against the DHCP static lease on the Master Server. Update the DHCP reservation if the MAC has changed before connecting the unit to VLAN 33.
- Configure the replacement unit with the IP address, hostname, MQTT credentials, and LED settings from the commissioning database before powering it on the network.
- For WLED controllers, push all presets from the Master Server backup to the replacement unit and verify cabin mode response before closing out the maintenance task (§3.4.4).
- Apply a new label to the replacement unit before installation. The label must carry the replacement unit's own S/N, P/N, HW REV, firmware version, and the current installation date. Do not reuse the removed unit's label.
- Record the replacement event in the aircraft maintenance log with the removed unit's S/N and the installed unit's S/N, P/N, HW REV, and firmware version.

⚠ REVIEW REQUIRED: Need feedback on MRI vs QR creation.

9. MOSAIC Open Architecture Display (MOAD)

The MOSAIC Open Architecture Display (MOAD) will describe an HTML5 template that can be used to create graphical user interfaces and passenger interfaces for touchscreen panels. It is intended to be similar in concept to open web-component libraries.

⚠ TO BE DEFINED: This section is to be defined in a future revision.

Appendix A — OpenWRT VLAN Configuration Example

The following is a reference configuration for an OpenWRT-based Master Router implementing the MOSAIC VLAN architecture. OpenWRT uses the DSA (Distributed Switch Architecture) network model on hardware shipped with OpenWRT 21.02 or later. All configuration files reside in `/etc/config/` and are applied with the `'service network restart'` and `'service firewall restart'` commands.

NOTE: This example covers a single physical router with multiple LAN ports. In deployments using a separate managed switch, the switch must be configured to carry the same 802.1Q VLAN tags on its trunk uplink to the router. The router port connecting to the switch must be configured as a tagged trunk port for all MOSAIC VLANs.

A.1 Required Packages

Install the following packages on a fresh OpenWRT image before applying the configuration:

```
opkg update
opkg install kmod-vlan-8021q kmod-bridge luci-app-firewall mosquitto-nossl
```

A.2 Network Configuration (`/etc/config/network`)

Define a VLAN-filtering bridge, one bridge-vlan stanza per ATA chapter, and a static interface (SVI) per VLAN. The example below shows three VLANs (15, 33, 44) and can be extended for the full set.

```
# — Bridge device with VLAN filtering —————
config device
    option name      'br-mosaic'
    option type      'bridge'
    option bridge_empty '1'
    list  ports      'lan1'      # trunk to distribution switch (tagged)
    list  ports      'lan2'      # direct connection to CMS server
    list  ports      'lan3'      # direct connection to lighting segment
    option vlan_filtering '1'

# — VLAN 15 — Core Network (tagged on trunk port lan1) —————
config bridge-vlan
    option device 'br-mosaic'
    option vlan   '15'
    list  ports  'lan1:t'
    list  ports  'lan2:t'

# — VLAN 16 — General Passenger Network (tagged on trunk port lan1) —
config bridge-vlan
    option device 'br-mosaic'
    option vlan   '16'
    list  ports  'lan1:t'
```

```
# — VLAN 17 — Crew Network (tagged on trunk port lan1) —————
config bridge-vlan
    option device 'br-mosaic'
    option vlan '17'
    list ports 'lan1:t'
```

```
# — VLAN 33 — Lighting (tagged trunk; untagged on lighting port) ———
config bridge-vlan
    option device 'br-mosaic'
    option vlan '33'
    list ports 'lan1:t'
    list ports 'lan3:u*' # untagged default VLAN for lighting port
```

```
# — VLAN 44 — CMS / IFE —————
config bridge-vlan
    option device 'br-mosaic'
    option vlan '44'
    list ports 'lan1:t'
    list ports 'lan2:u*'
```

```
# Repeat config bridge-vlan blocks for VLANs 21, 23, 24, 25, 26
```

```
# — VLAN Interfaces (SVIs — one gateway per VLAN) —————
```

```
config interface 'core_net'
    option device 'br-mosaic.15'
    option proto 'static'
    option ipaddr '10.15.0.1'
    option netmask '255.255.0.0'
```

```
config interface 'pax_net'
    option device 'br-mosaic.16'
    option proto 'static'
    option ipaddr '10.16.0.1'
    option netmask '255.255.0.0'
```

```
config interface 'crew_net'
    option device 'br-mosaic.17'
    option proto 'static'
    option ipaddr '10.17.0.1'
    option netmask '255.255.0.0'
```

```
config interface 'lighting'
    option device 'br-mosaic.33'
    option proto 'static'
    option ipaddr '10.33.0.1'
    option netmask '255.255.0.0'
```

```
config interface 'cms_ife'
    option device 'br-mosaic.44'
```

```
option proto    'static'
option ipaddr   '10.44.0.1'
option netmask  '255.255.0.0'
```

```
# No 'wan' interface is defined — WAN isolation enforced by absence.
```

A.3 DHCP Configuration (/etc/config/dhcp)

Each VLAN interface gets its own DHCP pool. Lighting controllers and other LRUs should be assigned static leases by MAC address to ensure IP stability across reboots.

```
config dhcp 'lighting'
    option interface 'lighting'
    option start     '20'
    option limit     '234'          # 10.33.0.20 - 10.33.0.253 - holding pool for
unreserved devices only
    option leasetime '12h'
```

```
# Static lease for a WLED wash lighting controller — 10.33.YY.ZZZ, YY = 10 (§2.9.2)
config host
    option mac    'AA:BB:CC:11:22:33'
    option ip     '10.33.10.103'
    option name   'wled-zone-forward-cove'
```

```
config dhcp 'cms_ife'
    option interface 'cms_ife'
    option start     '10'
    option limit     '500'
    option leasetime '12h'
```

```
config dhcp 'core_net'
    option interface 'core_net'
    option start '20'          # Core devices use static leases; small holding pool
only
    option limit '234'          # 10.15.0.20 - 10.15.0.253
    option leasetime '4h'
```

```
# Core infrastructure static leases
```

```
config host
    option mac    'AA:BB:CC:00:00:01'
    option ip     '10.15.0.2'
    option name   'mqtt-broker'
```

```
# VLAN 16 — general passenger network pool
```

```
config dhcp 'pax_net'
    option interface 'pax_net'
    option start     '10'
    option limit     '60000'      # 10.16.0.10 onwards — passenger device pool
    option leasetime '4h'
```

```
# VLAN 17 – crew network pool
config dhcp 'crew_net'
    option interface 'crew_net'
    option start '10'
    option limit '500'          # 10.17.0.10 onwards – crew device pool
    option leasetime '12h'
```

A.4 Firewall Configuration (/etc/config/firewall)

OpenWRT firewall zones map to network interfaces. Default policy for all cabin zones is INPUT DROP / OUTPUT ACCEPT / FORWARD DROP. Explicit forwarding rules open permitted flows between zones. No WAN zone is defined.

```
# — Zone definitions —————
config zone
    option name      'core_net'
    option input     'DROP'          # Explicit rules allow only required services
    option output    'ACCEPT'
    option forward   'DROP'
    list network    'core_net'
# Allow DHCP, DNS, NTP on VLAN 15 – core network devices
config rule
    option name      'core_services'
    option src       'core_net'
    option dest_port '53 67 68 123'
    option target    'ACCEPT'
# Restrict router management (SSH, LuCI) to 10.15.0.0/24 only
config rule
    option name      'mgmt_access'
    option src       'core_net'
    option src_ip    '10.15.0.0/24'  # Management devices only – first /24 of VLAN
15
    list dest_port  '22 80 443'
    option target    'ACCEPT'

# VLAN 16 – general passenger network: DHCP, DNS, NTP to the router only
config zone
    option name      'pax_net'
    option input     'DROP'
    option output    'ACCEPT'
    option forward   'DROP'
    list network    'pax_net'

config rule
    option name      'pax_services'
    option src       'pax_net'
    option dest_port '53 67 68 123'
    option target    'ACCEPT'
```

```
# VLAN 17 – crew network: DHCP, DNS, NTP to the router only
```

```
config zone
    option name 'crew_net'
    option input 'DROP'
    option output 'ACCEPT'
    option forward 'DROP'
    list network 'crew_net'

config rule
    option name 'crew_services'
    option src 'crew_net'
    option dest_port '53 67 68 123'
    option target 'ACCEPT'
```

```
config zone
    option name      'lighting'
    option input     'DROP'
    option output    'ACCEPT'
    option forward   'DROP'
    list network    'lighting'
```

```
config zone
    option name      'cms_ife'
    option input     'DROP'
    option output    'ACCEPT'
    option forward   'DROP'
    list network    'cms_ife'
```

```
# — Inter-zone forwarding rules —————
# Core network (VLAN 15) has full access to all cabin zones
config forwarding
    option src      'core_net'
    option dest     'lighting'
```

```
config forwarding
    option src      'core_net'
    option dest     'cms_ife'
```

```
# CMS <-> Lighting (bidirectional)
config forwarding
    option src      'cms_ife'
    option dest     'lighting'
```

```
config forwarding
    option src      'lighting'
    option dest     'cms_ife'
# Passenger network (VLAN 16) – CMS/IFE access and lighting API only
config forwarding
```

```

    option src 'pax_net'
    option dest 'cms_ife'

config rule
    option name 'pax_to_lighting_api'
    option src 'pax_net'
    option dest 'lighting'
    list proto 'tcp'
    option dest_port '80 443 1883'
    option target 'ACCEPT'

# Crew network (VLAN 17) – CMS/IFE access and lighting API only
config forwarding
    option src 'crew_net'
    option dest 'cms_ife'

config rule
    option name 'crew_to_lighting_api'
    option src 'crew_net'
    option dest 'lighting'
    list proto 'tcp'
    option dest_port '80 443 1883'
    option target 'ACCEPT'

# Passenger (VLAN 16) <-> Core (VLAN 15): no forwarding is defined here.
# Add explicit rules only as the program design and FAA ASISP
# certification permit (§2.9.3).

# Fire protection – read-only HTTP from any zone
config rule
    option name      'any_to_fire_readonly'
    option src       '*'
    option dest      'fire_protect'
    list proto      'tcp'
    list dest_port   '80'
    option target    'ACCEPT'

# All other cross-zone traffic is dropped by default zone policy.
# No WAN zone – no external routing path exists.
```

A.5 Applying the Configuration

```
# Restart networking to apply VLAN bridge and interface config
/etc/init.d/network restart
```

```
# Restart firewall to apply zone and forwarding rules
/etc/init.d/firewall restart
```

```
# Verify VLAN interfaces are up
ip addr show br-mosaic.33
ip addr show br-mosaic.44
```

```
# Test inter-VLAN connectivity (CMS to Lighting)
ping -I br-mosaic.44 10.33.10.103
```

```
# Verify no WAN route exists
ip route | grep default    # should return nothing
```

NOTE: LuCI (the OpenWRT web GUI) can be used to view and verify VLAN and firewall state after applying configuration. Navigate to Network > Interfaces and Network > Firewall to confirm all zones and rules are active.

Appendix B — Standard API Reference for General Status

All MOSAIC LRUs that expose an HTTP interface must implement the following standard API endpoints. These endpoints are in addition to any device-specific API (e.g., WLED JSON API for RGB controllers). Authentication is required on all endpoints per §2.9.11.

B.1 Base URL

All standard MOSAIC API endpoints are served from the base path `/api/v1/` on port 80 (HTTP) or 443 (HTTPS with TLS). Example base URL for a lighting controller: `http://10.33.10.163/api/v1/`

B.2 Standard Endpoints

Method	Endpoint	Auth Required	Description
GET	<code>/api/v1/health</code>	No	Lightweight liveness check. Returns HTTP 200 with <code>{"status":"ok"}</code> when the LRU is reachable and firmware is running. No-auth to allow unauthenticated polling by the Master Server health agent.
GET	<code>/api/v1/status</code>	Yes	Full LRU status object. Returns identity, power state, firmware version, uptime, network info, and active fault list. See B.3 for response schema.
GET	<code>/api/v1/faults</code>	Yes	Active fault log. Returns array of current faults: <code>[{"pri":"P3","code":"MQTT_DISCONNECT","ts":"2026-05-15T14:00:00Z","msg":"Broker unreachable"}]</code> . Empty array = no active faults.
POST	<code>/api/v1/faults/clear</code>	Yes	Clear all resolved faults from the active fault list. Body: <code>{"cleared_by":"maintenance-agent"}</code> . Returns updated fault array.
GET	<code>/api/v1/info</code>	Yes	Device identity and configuration: LRU type, ATA chapter, zone, MAC address, assigned IP, firmware build date, hardware revision.
GET	<code>/api/v1/network</code>	Yes	Network interface status: IP address, subnet, gateway, MQTT broker connection state, Ethernet link state and speed, packet counters.
GET	<code>/api/v1/log</code>	Yes	Recent event log entries. Query parameters: <code>?limit=50</code> (max 200), <code>?pri=P1</code> (filter by priority), <code>?since=<ISO timestamp></code> . Returns array of log objects.
POST	<code>/api/v1/reboot</code>	Yes	Command a controlled reboot. Body: <code>{"reason":"maintenance"}</code> . LRU publishes an MQTT event before rebooting. Use during maintenance windows only.

GET	/api/v1/ping	No	Echo endpoint. Returns {"ts":"<UTC>","id":"<lru-id>"}. Used by Master Server to measure round-trip latency and confirm NTP sync.
-----	--------------	----	--

B.3 /api/v1/status Response Schema

The /api/v1/status endpoint returns a JSON object with the following required fields. LRU-specific fields (e.g., bri, eth_link for WLED) are included in the optional ext object.

```
{
  "id":      "wled-zone-fwd-cove",
  "ata":     33,
  "zone":    "forward-cove",
  "ip":      "10.33.10.163",
  "mac":     "AA:BB:CC:11:22:33",
  "fw":      "0.15.0",
  "hw_rev":  "1.2",
  "uptime_s": 86400,
  "power":   "on",
  "status":  "ok",
  "faults":  [],
  "ts":      "2026-05-15T14:32:00Z",
  "ext": {
    "bri":    180,
    "eth_link": true,
    "free_heap": 32768,
    "leds":   144
  }
}
```

Field	Type	Description
id	string	Human-readable LRU identifier. Must match MQTT <lru-id> in topic path and DHCP hostname.
ata	integer	ATA chapter number (e.g. 33 for lighting). Matches VLAN ID.
zone	string	Lighting zone or subsystem label (e.g. forward-cove, galley-wash).
ip	string	LRU's current IP address.
mac	string	LRU MAC address. Used for DHCP static lease validation.
fw	string	Firmware version string. Compared against Master Server repository for compliance.
hw_rev	string	Hardware revision. Logged for maintenance and spares tracking.
uptime_s	integer	Seconds since last boot. Unexpected resets (<300 s uptime) logged as P3.
power	string	on or standby. Reflects commanded and actual power state.

status	string	ok, degraded, or fault. Derived from faults array.
faults	array	Active fault objects. Empty = healthy. Each fault: {pri, code, ts, msg}.
ts	string	ISO 8601 UTC timestamp of the status response. Used to detect stale responses.
ext	object	Optional LRU-type-specific fields. WLED: bri, eth_link, free_heap, leds. CMS: cpu_pct, disk_free_mb.

B.4 HTTP Status Codes

HTTP Code	Meaning	MOSAIC Handling
200 OK	Request succeeded	Normal response. Parse JSON body.
204 No Content	Command accepted, no body	Used for POST commands (reboot, clear). No body to parse.
400 Bad Request	Malformed request body or params	Log error. Check payload format. Do not retry automatically.
401 Unauthorized	Missing or invalid API key/token	Log as P3. Verify credentials on Master Server. Re-provision if needed.
404 Not Found	Endpoint not implemented	LRU does not support this endpoint. Check firmware version and interface manifest.
503 Unavailable	LRU busy or in maintenance mode	Retry after 30 s. If persists >120 s, log as P3.
Timeout	No TCP response within 5 s	Log as P3 Communication Failure. Increment miss counter; three consecutive misses = P1.

B.5 Authentication

All authenticated endpoints (/api/v1/status, /faults, /info, /network, /log, /reboot) require an API key passed in the HTTP Authorization header. The Master Server stores and rotates API keys per LRU during commissioning.

```
Authorization: Bearer <api-key>
```

- API keys must be unique per LRU and stored in the Master Server's commissioning database.
- Keys must be rotated at least once per maintenance cycle or whenever a unit is replaced.
- The /api/v1/health and /api/v1/ping endpoints are intentionally unauthenticated to allow passive health polling without credential management overhead.
- OAuth 2.0 client credentials flow is the preferred authentication mechanism for future revisions (§2.9.11).

References

Sources for §2.1 Architecture

- Office of the Under Secretary of Defense for Research and Engineering, Modular Open Systems Approach (MOSA) Reference Frameworks in Defense Acquisition Programs, May 2020. The definitions and implementation steps in §2.1 are drawn from this document (Table 2-2 and §2.4.2), which cites the two works below. <https://ac.cto.mil/wp-content/uploads/2020/06/MOSA-Ref-Framework-May2020.pdf>
- Firesmith and Donohoe 2015, as cited in the document above: Firesmith, Donald, and Pat Donohoe. 2015. Model Open System Architecture (OSA) Requirements. Special Report, Pittsburgh: Software Engineering Institute, Carnegie Mellon University.
- Langlois 2002, as cited in the document above (where the name is printed “Lanlois”): Langlois, Richard N. 2002. “Modularity in Technology and Organization.” *Journal of Economic Behavior & Organization*, pp. 19–37.

Modular Open Systems Approach (MOSA)

- 10 U.S.C. § 4401, Requirement for modular open system approach in major defense acquisition programs; definitions. <https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title10-section4401&num=0&edition=prelim>
- U.S. Department of Defense, Implementing a Modular Open Systems Approach in Department of Defense Programs (MOSA Implementation Guidebook), February 2025. <https://www.cto.mil/wp-content/uploads/2025/03/MOSA-Implementation-Guidebook-27Feb2025-Cleared.pdf>
- Secretaries of the Army, Navy, and Air Force, Tri-Service Memorandum, “Modular Open Systems Approaches for our Weapon Systems is a Warfighting Imperative,” 2019; updated December 2024.
- Office of the Under Secretary for Research and Engineering, Modular Open Systems Approach resource page. <https://www.cto.mil/sea/mosa/>

Sensor Open Systems Architecture (SOSA)

- The Open Group SOSA Consortium, Technical Standard for SOSA Reference Architecture. Clause numbers cited in this document as [SOSA n.n] refer to this standard. <https://www.opengroup.org>
- The Open Group, Modular Open Systems Approach (MOSA) overview and related open standards (SOSA, FACE). <https://www.opengroup.org/face/mosa>

Wireless (cited in §2.9.10)

- Hall, Colby. Passenger Wireless in VIP Aircraft Cabins: Design Constraints, Material Interference, and Network Architecture for WLAN in VIP Environments. Technical White Paper, Aloft Aeroarchitects, February 2023.

Video

- HDMI Licensing, LLC, HDMI 2.0 Specification press release (4K@60 at 18 Gbps). <https://www.hdmi.org/press/bodydetails/20>

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For questions contact the MOSAIC Working Group